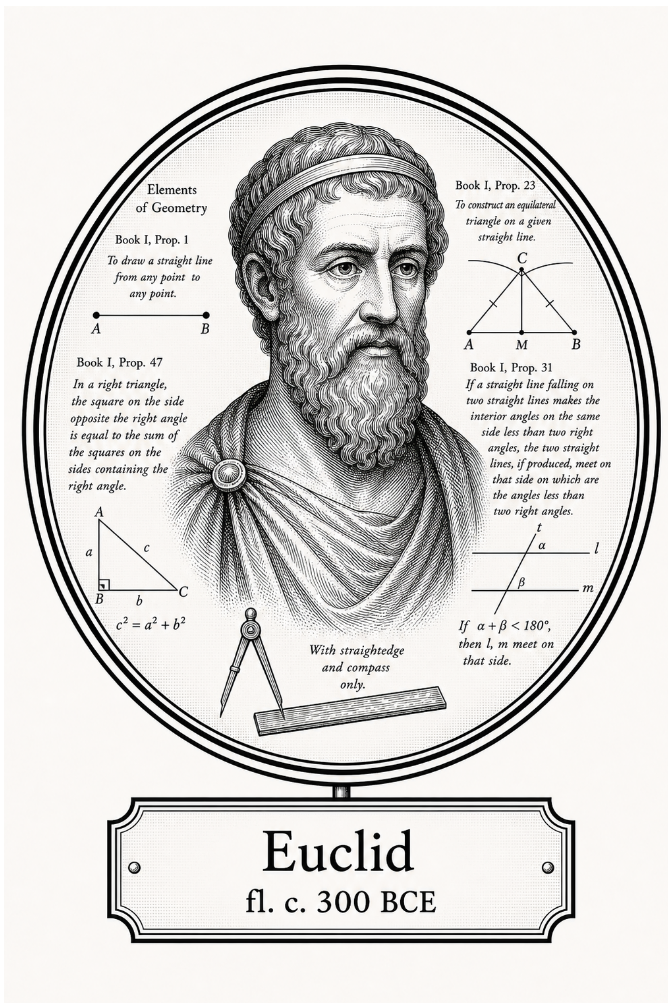
From Cantor to Ito

Logic, Sets, and Proof

Foundational Geometry



Euclid

fl. c. 300 BCE

Self-Study Notes

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July 2, 2026

For every reader learning to make mathematics their own.

Contents

Chapter 1

Euclidean Geometry

SERIES	VOLUME I	BOOK III	CHAPTER 10
From Cantor to Ito	Logic, Sets, and Proof	Foundational Geometry	Euclidean Geometry

1.1 Foundations

Definitions

Remark (Source note). The following definitions are Euclid's Book I definitions in the Heath translation [Euclid, 1956, Book I, Definitions 1–23].

Definition

Definition 1.1.1 (Point)**.** A point is that which has no part.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.2 (Line)**.** A line is breadthless length.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.3 (Ends of a line)**.** The ends of a line are points.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.4 (Straight line)**.** A straight line is a line which lies evenly with the points on itself.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.5 (Surface)**.** A surface is that which has length and breadth only.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.6 (Edges of a surface)**.** The edges of a surface are lines.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.7 (Plane surface)**.** A plane surface is a surface which lies evenly with the straight lines on itself.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.8 (Plane angle)**.** A plane angle is the inclination to one another of two lines in a plane which meet one another and do not lie in a straight line.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.9 (Rectilinear angle)**.** And when the lines containing the angle are straight, the angle is called rectilinear.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.10 (Right angle and perpendicular)**.** When a straight line standing on a straight line makes the adjacent angles equal to one another, each of the equal angles is right, and the straight line standing on the other is called a perpendicular to that on which it stands.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.11 (Obtuse angle)**.** An obtuse angle is an angle greater than a right angle.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.12 (Acute angle)**.** An acute angle is an angle less than a right angle.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.13 (Boundary)**.** A boundary is that which is an extremity of anything.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.14 (Figure)**.** A figure is that which is contained by any boundary or boundaries.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.15 (Circle)**.** A circle is a plane figure contained by one line such that all the straight lines falling upon it from one point among those lying within the figure equal one another.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.16 (Center of a circle)**.** And the point is called the center of the circle.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.17 (Diameter)**.** A diameter of the circle is any straight line drawn through the center and terminated in both directions by the circumference of the circle, and such a straight line also bisects the circle.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.18 (Semicircle)**.** A semicircle is the figure contained by the diameter and the circumference cut off by it. And the center of the semicircle is the same as that of the circle.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.19 (Rectilinear figures)**.** Rectilinear figures are those which are contained by straight lines, trilateral figures being those contained by three, quadrilateral those contained by four, and multilateral those contained by more than four straight lines.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.20 (Equilateral, isosceles, and scalene triangles)**.** Of trilateral figures, an equilateral triangle is that which has its three sides equal, an isosceles triangle that which has two of its sides alone equal, and a scalene triangle that which has its three sides unequal.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.21 (Right-angled, obtuse-angled, and acute-angled triangles)**.** Further, of trilateral figures, a right-angled triangle is that which has a right angle, an obtuse-angled triangle that which has an obtuse angle, and an acute-angled triangle that which has its three angles acute.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.22 (Quadrilateral classes)**.** Of quadrilateral figures, a square is that which is both equilateral and right-angled; an oblong that which is right-angled but not equilateral; a rhombus that which is equilateral but not right-angled; and a rhomboid that which has its opposite sides and angles equal to one another but is neither equilateral nor right-angled. And let quadrilaterals other than these be called trapezia.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Definition

Definition 1.1.23 (Parallel straight lines)**.** Parallel straight lines are straight lines which, being in the same plane and being produced indefinitely in both directions, do not meet one another in either direction.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Postulates

Remark (Source note). The following are Euclid's five postulates from Book I of the *Elements* in the Heath translation [Euclid, 1956, Book I, Postulates 1–5].

Remark (Exposition). Postulates are geometry-specific starting assumptions. They say which constructions are permitted and which basic geometric facts may be used without proof. In Euclid's system, these are not conclusions reached from earlier results; they are part of the foundation from which the theory begins.

Later propositions are proved from the definitions, the postulates, the common notions, and propositions already established. Thus the postulates are the rules of play for Euclidean geometry: they determine what can be constructed, what may be assumed about straight lines and circles, and where the theory's geometric content enters before proof begins.

Axiom

Axiom 1.1.24 (Euclid Postulate I: Drawing a straight line)**.** Let it have been postulated to draw a straight line from any point to any point.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 1.1.25 (Euclid Postulate II: Producing a finite straight line)**.** Let it have been postulated to produce a finite straight line continuously in a straight line.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 1.1.26 (Euclid Postulate III: Describing a circle)**.** Let it have been postulated to describe a circle with any center and distance.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 1.1.27 (Euclid Postulate IV: Equality of right angles)**.** Let it have been postulated that all right angles are equal to one another.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 1.1.28 (Euclid Postulate V: Parallel postulate)**.** Let it have been postulated that, if a straight line falling on two straight lines makes the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side on which are the angles less than the two right angles.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axioms

Remark (Source note)*.* Euclid calls the following principles *common notions.* They function as general axioms governing equality and comparison in Book I of the *Elements* [Euclid, 1956, Book I, Common Notions 1–5].

Axiom

Axiom 1.1.29 (Euclid Common Notion I: Equality is transitive through a common equal)**.** Things which are equal to the same thing are also equal to one another.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 1.1.30 (Euclid Common Notion II: Adding equals to equals)**.** If equals be added to equals, the wholes are equal.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 1.1.31 (Euclid Common Notion III: Subtracting equals from equals)**.** If equals be subtracted from equals, the remainders are equal.

Remark (Standard quantified statement)*.* This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 1.1.32 (Euclid Common Notion IV: Coincidence implies equality)**.** Things which coincide with one another are equal to one another.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Axiom

Axiom 1.1.33 (Euclid Common Notion V: Whole and part)**.** The whole is greater than the part.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Remark (Exposition). Common notions are general principles of reasoning rather than specifically geometric assumptions. They express basic behavior of equality, comparison, and logical transfer: equal things may be substituted for equal things, adding or subtracting equals preserves equality, and a whole exceeds its part.

Because these principles are not tied to points, lines, circles, or planes, they can be used throughout mathematics, not only in geometry. In modern language, they play the role of background axioms governing how equality and comparison may be used inside proofs.

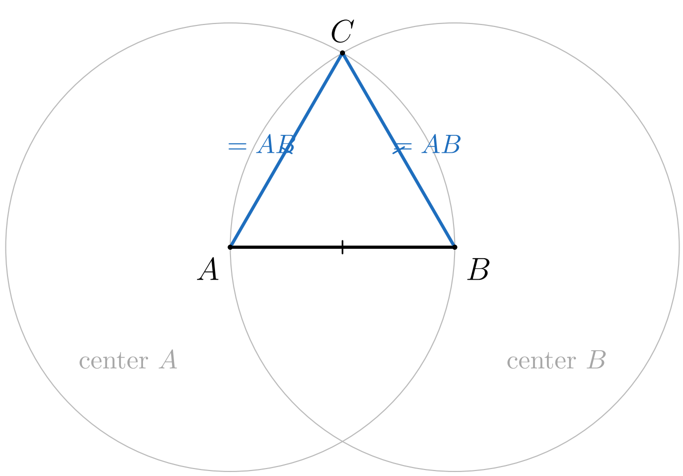
1.2 Compass-and-Straightedge Constructions

Propositions

Remark (Source note). The following propositions are selected from Euclid's Book I in modern theorem form, following the Heath translation of the *Elements* [Euclid, 1956, Book I]. The selection is curated for the trigonometric construction: congruence, parallel-angle control, triangle angle-sum, area comparison, square construction, and the Pythagorean theorem.

Theorem

Theorem 1.2.1 (Euclid I.1: Constructing an equilateral triangle)**.** *Given a finite straight line segment AB, there exists an equilateral triangle ABC constructed on AB.*

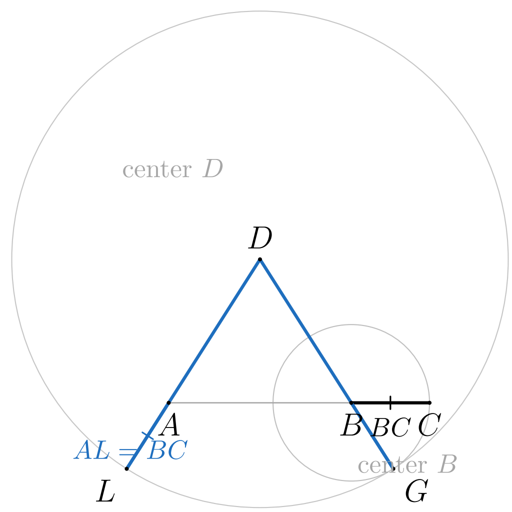


Remark (Proof). Go to proof.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Theorem

Theorem 1.2.2 (Euclid I.2: Copying a segment from a given point)**.** *Given a point A and a finite straight line segment BC, there exists a straight line segment beginning at A and equal to BC.*

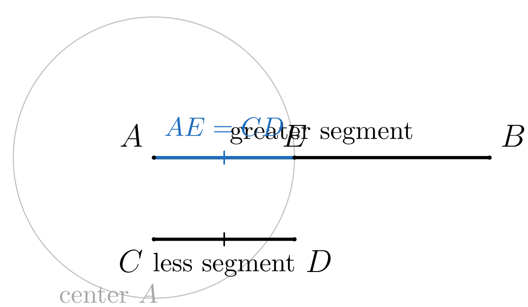


Remark (Proof). Go to proof.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Theorem

Theorem 1.2.3 (Euclid I.3: Cutting off an equal segment)**.** *Given two unequal finite straight line segments, there exists a construction which cuts off from the greater a straight line segment equal to the less.*

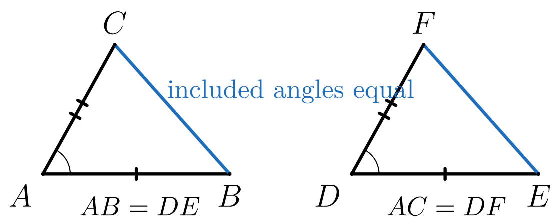


Remark (Proof). Go to proof.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Theorem

Theorem 1.2.4 (Euclid I.4: Side-angle-side congruence)**.** *If two triangles have two sides equal to two sides respectively, and have the angles contained by the equal straight lines equal, then they also have the base equal to the base, the triangle equal to the triangle, and the remaining angles equal to the remaining angles respectively.*

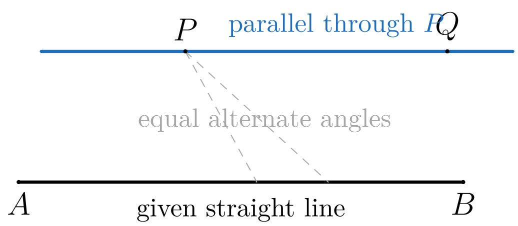


Remark (Proof). Go to proof.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Theorem

Theorem 1.2.5 (Euclid I.31: Drawing a parallel through a point)**.** *Through a given point, to draw a straight line parallel to a given straight line.*

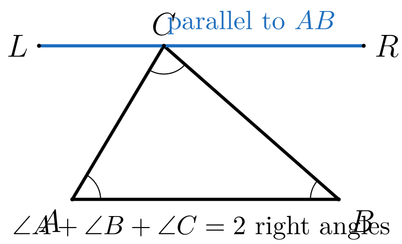


Remark (Proof). Go to proof.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Theorem

Theorem 1.2.6 (Euclid I.32: Triangle angle sum)**.** *In any triangle, if one of the sides is produced, then the exterior angle is equal to the two interior and opposite angles, and the three interior angles of the triangle are equal to two right angles.*

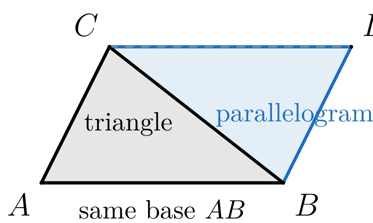


Remark (Proof). Go to proof.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Theorem

Theorem 1.2.7 (Euclid I.41: Triangle and parallelogram on the same base). *If a parallelogram has the same base with a triangle and is in the same parallels, then the parallelogram is double the triangle.*

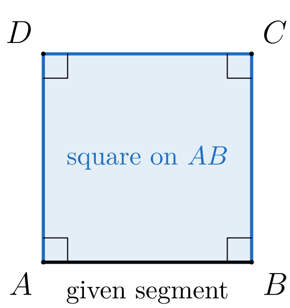


Remark (Proof). Go to proof.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Theorem

Theorem 1.2.8 (Euclid I.46: Constructing a square on a segment). *On a given straight line, to describe a square.*

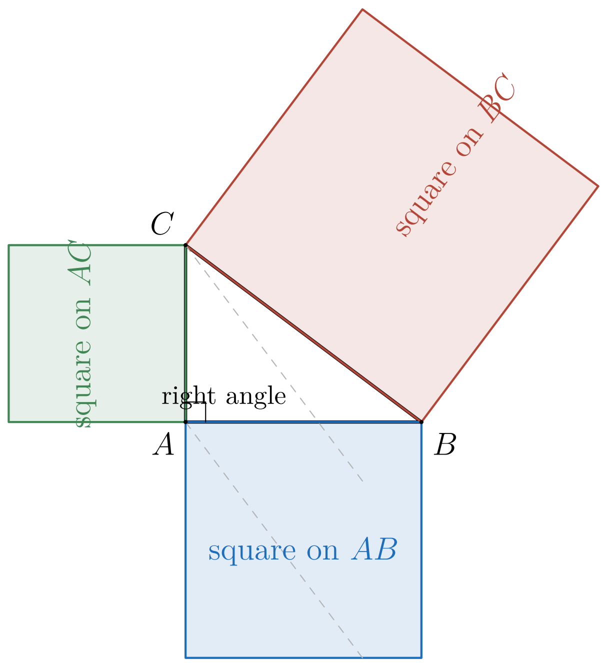


Remark (Proof). Go to proof.

Remark (Standard quantified statement). This is a classical Euclidean source-text clause. In this chapter it is treated as a named primitive statement rather than expanded into a modern first-order formalization.

Theorem

Theorem 1.2.9 (Euclid I.47: Pythagorean theorem)**.** *In right-angled triangles, the square on the side subtending the right angle is equal to the squares on the sides containing the right angle.*



Remark (Proof). Go to proof.

Remark (Placement). This proposition is pulled forward as the Book I capstone needed by the trigonometric definitions. Its professional proof depends on later Book I propositions that are not yet developed in this chapter; the proof file records those pending Euclidean nodes explicitly.

Proofs

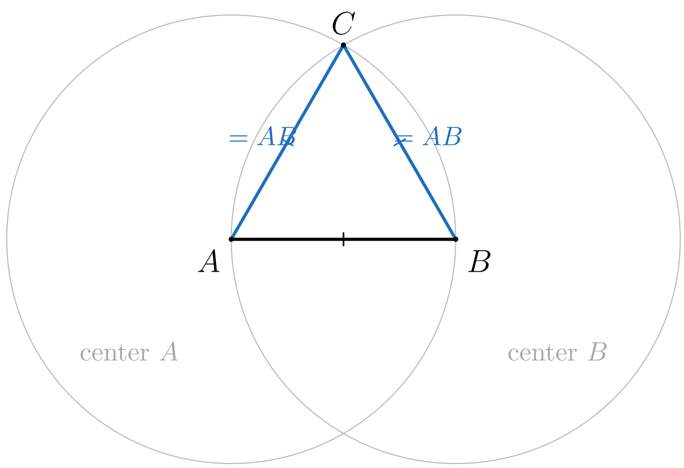
Remark (Return). Return to Theorem

Theorem (Euclid I.1: Constructing an equilateral triangle)**.** *Given a finite straight line segment AB, there exists an equilateral triangle ABC constructed on AB.*

Professional Standard Proof.
Let AB be the given finite straight line. Describe the circle with center A and radius AB, and describe the circle with center B and radius BA. Let C be an intersection point of the two circles, and draw the straight lines CA and CB.

Since C lies on the circle with center A, $AC = AB$. Since C lies on the circle with center B, $BC = BA$. Thus AC and BC are each equal to AB, so by Common Notion I they are equal to one another. Therefore AB, BC, and CA are equal, and the triangle ABC is equilateral. □

Detailed Learning Proof.



The construction uses exactly the permitted tools available at this point. Postulate I allows the given segment AB to be treated as a drawable straight line segment. Postulate III allows a circle with any center and distance, so we may draw one circle centered at A with distance AB and another centered at B with distance BA.

Choose an intersection point C of the two circles. The segment AC is a radius of the circle centered at A, so it equals the radius AB. The segment BC is a radius of the circle centered at B, so it equals the radius BA, which is the same segment as AB. Hence all three sides of $\triangle ABC$ are equal, and by the definition of an equilateral triangle, $\triangle ABC$ is equilateral. □

Remark (Proof structure). Construct two circles from the endpoints of the given segment, draw the two radii to an intersection point, and use equality through the common segment AB to identify the three sides.

Remark (Euclid's intersection gap). The proof assumes that the two circles meet. This is the classical lacuna in Euclid I.1: the postulates permit drawing the circles, but Euclid's stated axioms do not by themselves include a circle–circle intersection principle. Later axiomatizations add continuity or incidence assumptions that justify this step.

Remark (Dependencies).

- Point
- Straight line
- Circle
- Equilateral triangle
- Euclid Postulate I
- Euclid Postulate III
- Euclid Common Notion I

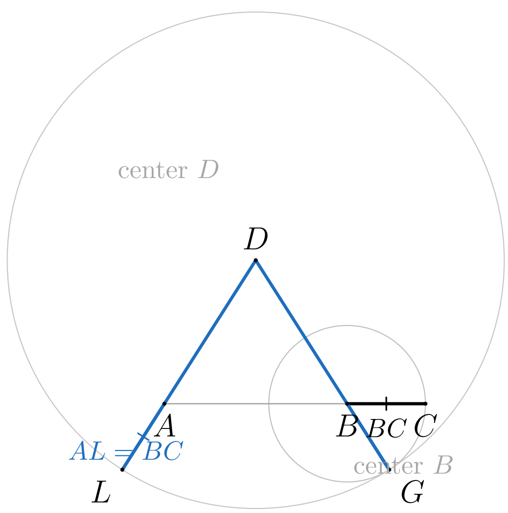
Remark (Return). Return to Theorem

Theorem (Euclid I.2: Copying a segment from a given point)**.** *Given a point A and a finite straight line segment BC, there exists a straight line segment beginning at A and equal to BC.*

Professional Standard Proof.
Let A be the given point and let BC be the given segment. Join AB and construct an equilateral triangle DAB on AB by Euclid I.1. Produce the straight lines DA and DB. With center B and distance BC, describe a circle meeting the produced line DB at G. With center D and distance DG, describe a circle meeting the produced line DA at L.

Since $BG = BC$ and $DL = DG$, while $DA = DB$ because DAB is equilateral, the remainders AL and BG are equal. Therefore $AL = BC$. Thus from the point A a straight line segment equal to the given segment BC has been constructed. □

Detailed Learning Proof.



The role of the equilateral triangle is to create two matching initial lengths: $DA = DB$. After the side DB is extended, the circle centered at B marks a point G with $BG = BC$. Then the circle centered at D transfers the full length DG onto the other produced side, giving $DL = DG$.

Now compare the two long segments from D. On one side, $DG = DB + BG$; on the other, $DL = DA + AL$. Since $DG = DL$ and $DB = DA$, subtracting the equal initial parts leaves $BG = AL$. But $BG = BC$, so $AL = BC$ by Common Notion I. The segment AL begins at the required point A and has the required length. □

Remark (Proof structure). Euclid I.2 is a transfer construction. It does not assume a compass can be lifted and moved directly; instead it copies the length BC using only straight lines, produced straight lines, circles, Euclid I.1, and common notions about equality.

Remark (Dependencies).

- Point

- Straight line
- Circle
- Euclid I.1
- Euclid Postulate I
- Euclid Postulate II
- Euclid Postulate III
- Euclid Common Notion I

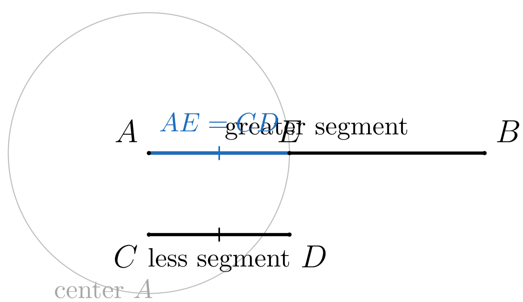
Remark (Return). Return to Theorem

Theorem (Euclid I.3: Cutting off an equal segment)**.** *Given two unequal finite straight line segments, there exists a construction which cuts off from the greater a straight line segment equal to the less.*

Professional Standard Proof.
Let AB be the greater of the two given unequal segments and let CD be the less. By Euclid I.2, construct from the point A a segment AD equal to CD. With center A and distance AD, describe a circle meeting AB at E.

Since AE and AD are radii of the same circle, $AE = AD$. Since $AD = CD$ by construction, $AE = CD$ by Common Notion I. Therefore the segment AE has been cut off from the greater segment AB equal to the less segment CD. □

Detailed Learning Proof.



The problem is not merely to draw a segment equal to the shorter one; Euclid I.2 already does that. The problem is to place such a segment on the greater given segment. Starting at the endpoint A of the greater segment AB, copy the shorter segment CD to form AD. A circle centered at A with radius AD records exactly that copied length in every direction from A.

Where this circle meets the greater segment AB, call the point E. Then AE is a radius of the circle and so is equal to AD. Since AD was copied from CD, the cut-off segment AE is equal to CD. □

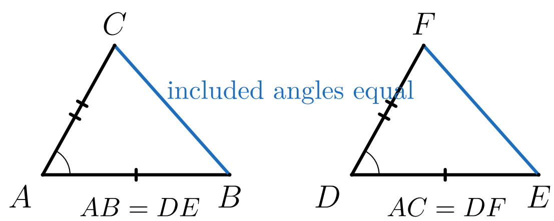
Remark (Proof structure)*.* Euclid I.3 depends on I.2 as a length-copying lemma. Once the shorter segment has been copied from the endpoint of the greater one, Postulate III turns that copied length into the cutoff point on the greater segment.

Remark (Dependencies)*.*

- Straight line
- Circle
- Euclid I.2
- Euclid Postulate III
- Euclid Common Notion I

Remark (Return). Return to Theorem

Theorem (Euclid I.4: Side-angle-side congruence)**.** *If two triangles have two sides equal to two sides respectively, and have the angles contained by the equal straight lines equal, then the bases, the triangles, and the remaining angles are equal respectively.*



Professional Standard Proof.
Let triangles ABC and DEF satisfy $AB = DE$, $AC = DF$, and $\angle BAC = \angle EDF$. Place the point A on D, the straight line AB on DE, and the straight line AC on DF. Since the included angles are equal, the two sides fall along the corresponding sides. Since $AB = DE$, the point B coincides with E; since $AC = DF$, the point C coincides with F.

Therefore the base BC coincides with the base EF. By Common Notion IV, things which coincide are equal. Hence $BC = EF$, the whole triangle ABC is equal to the whole triangle DEF, and the remaining corresponding angles are equal. □

Detailed Learning Proof.
The diagram is meant to be read by superposition. Equal included angles allow one triangle to be laid on the other without rotating one side away from its mate. The two equal side lengths then force the two non-vertex endpoints to land exactly on the corresponding endpoints. Once the three vertices coincide, the third side and the two remaining angles coincide as well. □

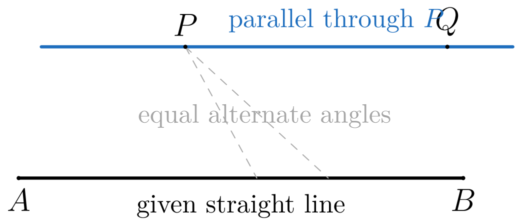
Remark (Proof structure). Superpose one triangle on the other, use the two equal sides to identify the endpoints, and use coincidence to conclude equality of the base and remaining angles.

Remark (Dependencies).

- Triangle side classes
- Rectilinear angle
- Euclid Common Notion IV

Remark (Return). Return to Theorem

Theorem (Euclid I.31: Drawing a parallel through a point)**.** *Through a given point, to draw a straight line parallel to a given straight line.*



Professional Standard Proof.
Let P be the given point and let AB be the given straight line. Draw a transversal from P to a point on AB. At P, construct an angle equal to the alternate interior angle made by the transversal with AB, and draw the straight line through P forming that equal angle.

The two lines cut by the transversal have equal alternate interior angles. Euclid's parallel criterion, which rests on the parallel postulate, gives that the constructed line through P is parallel to AB. □

Detailed Learning Proof.
The construction copies an angle rather than guessing a direction. Once the angle at P matches the alternate angle at the given line, the transversal sees the same slant at both intersections. In Euclidean geometry that equality of alternate interior angles is exactly the signal that the two straight lines do not meet when produced. □

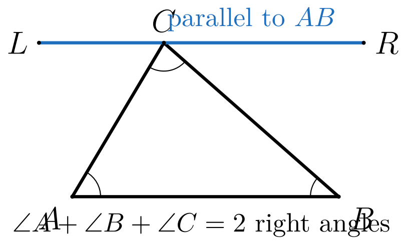
Remark (Proof structure). Draw a transversal, copy the alternate angle at the given point, and invoke the parallel criterion to identify the constructed line as parallel to the given line.

Remark (Dependencies).

- Parallel straight lines
- Euclid Postulate I
- Euclid Postulate V

Remark (Return). Return to Theorem

Theorem (Euclid I.32: Triangle angle sum)**.** *In any triangle, if one of the sides is produced, then the exterior angle is equal to the two interior and opposite angles, and the three interior angles of the triangle are equal to two right angles.*



Professional Standard Proof.
Let ABC be a triangle. Through C, draw a line parallel to AB by Euclid I.31. The straight line AC cuts the two parallels, so the angle made by AC with the parallel through C equals the interior angle at A. Likewise, the straight line BC cuts the two parallels, so the angle made by BC with the parallel through C equals the interior angle at B.

At the point C, these two copied angles together with the original angle ACB form a straight angle, that is, two right angles. Substituting the equal alternate angles for the angles at A and B, the three interior angles of triangle ABC are equal to two right angles. The same equal-angle comparison also identifies the exterior angle with the sum of the two opposite interior angles. □

Detailed Learning Proof.
The diagram places a parallel to the base through the top vertex. The base angle at A reappears at C on one side of the triangle, and the base angle at B reappears at C on the other side. Along the parallel line through C, those two copied angles and the original vertex angle exactly fill a straight line. A straight line has two right angles, so the three angles of the triangle sum to two right angles. □

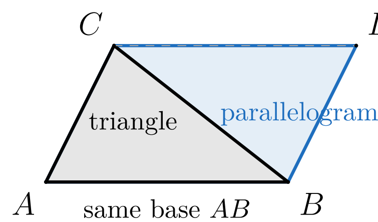
Remark (Proof structure). Draw a parallel through one vertex, transfer the two remote angles by alternate interior angles, and read the three angles as a straight angle.

Remark (Dependencies).

- Triangle angle classes
- Euclid I.31
- Euclid Postulate V

Remark (Return). Return to Theorem

Theorem (Euclid I.41: Triangle and parallelogram on the same base)**.** *If a parallelogram has the same base with a triangle and is in the same parallels, then the parallelogram is double the triangle.*



Professional Standard Proof.
Let parallelogram $ABDC$ and triangle ABC have the same base AB and lie between the same parallels. Draw the diagonal BC of the parallelogram. The diagonal divides the parallelogram into two triangles on equal bases and in the same parallels. These two triangles coincide in area by the earlier parallel-area comparison.

One of those two triangles is the given triangle ABC. Hence the whole parallelogram consists of two equal copies of that triangle, so the parallelogram is double the triangle. □

Detailed Learning Proof.
Look at the diagonal. It cuts the parallelogram into two triangular halves. Because opposite sides of the parallelogram are parallel, the two halves have matching bases and matching altitude between the same parallel lines. Euclid therefore treats them as equal in area. The parallelogram is the two halves together, while the given triangle is one half. □

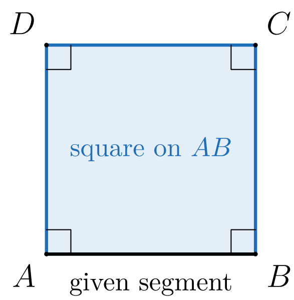
Remark (Proof structure). Cut the parallelogram by a diagonal, compare the two triangular halves, and read the whole parallelogram as two equal copies of the triangle.

Remark (Dependencies).

- Parallel straight lines
- Quadrilateral classes
- Euclid Common Notion IV

Remark (Return). Return to Theorem

Theorem (Euclid I.46: Constructing a square on a segment)**.** *On a given straight line, to describe a square.*



Professional Standard Proof.
Let AB be the given segment. At A, draw a perpendicular to AB and cut off AD equal to AB. Through D, draw a line parallel to AB; through B, draw a line parallel to AD. Let those two parallels meet at C.

The quadrilateral $ABCD$ has opposite sides parallel, so it is a parallelogram. The construction gives one right angle, and the parallel lines transfer right angles to the remaining corners. Also $AD = AB$ by construction, and opposite sides of the parallelogram are equal. Thus all four sides are equal and all four angles are right angles. By the definition of a square, $ABCD$ is the square on AB. □

Detailed Learning Proof.
The construction first creates one side perpendicular to the given segment and equal to it. The two parallels then close the quadrilateral without changing the directions: horizontal sides stay parallel to AB, and vertical sides stay parallel to AD. This makes a right-angled parallelogram whose adjacent sides are equal, hence a square. □

Remark (Proof structure). Construct a perpendicular equal side, close the figure with parallels, and use parallel transfer of right angles plus equality of opposite sides to identify a square.

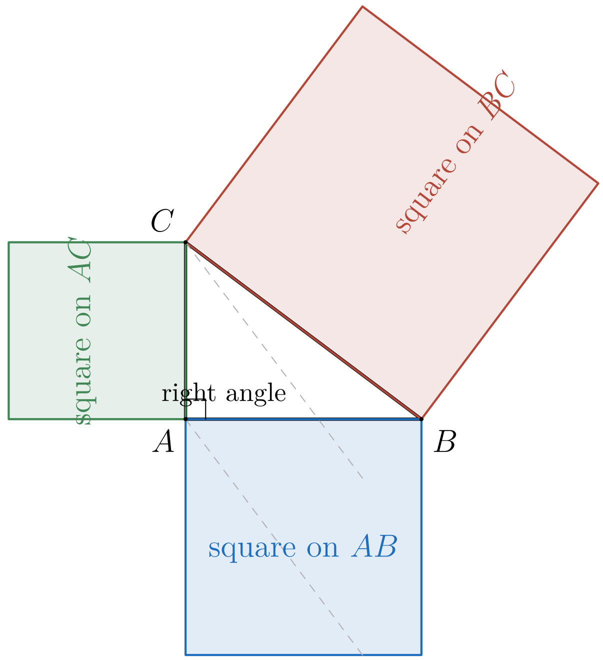
Remark (Dependencies).

- Square
- Euclid I.31
- Euclid Postulate I
- Euclid Postulate IV

Remark (Return). Return to Theorem

Theorem (Euclid I.47: Pythagorean theorem)**.** *In right-angled triangles, the square on the side subtending the right angle is equal to the squares on the sides containing the right angle.*

Professional Standard Proof.



Let ABC be a right-angled triangle with right angle at A. Construct the squares on AB, AC, and BC by Euclid I.46. Through A draw the line parallel to the appropriate sides of the square on BC, dividing that square into two rectangles.

By Euclid I.4, the triangles built from the side AB and the corresponding side of the hypotenuse-square are congruent in the required side-angle-side comparison. By Euclid I.41, the square on AB is double one of those triangles, and the matching rectangle in the hypotenuse-square is double the congruent triangle. Hence the square on AB equals that rectangle. The same argument on the side AC shows that the square on AC equals the other rectangle in the hypotenuse-square.

The two rectangles together make up the square on BC. Therefore, by Common Notion II, the square on AB together with the square on AC equals the square on BC. □

Detailed Learning Proof.
Euclid's proof is an area comparison, but not an algebraic computation of areas as real numbers. The square on the hypotenuse is split into two rectangles. One rectangle is shown equal to the square on one leg; the other rectangle is shown equal to the square on the other leg. Since the two rectangles together make up the square on the hypotenuse, the square on the hypotenuse equals the sum of the two leg-squares.

This matters for the later trigonometric construction. When a right triangle is normalized so that its hypotenuse has unit length, Euclid I.47 is the synthetic source of the identity

$$\cos^2\theta + \sin^2\theta = 1.$$

The coordinate and area-algebra version of this proof belongs later, after Cartesian coordinates and numerical area have been licensed. □

Remark (Dependency status). This is the Euclidean windmill proof at the level needed for the trigonometric construction. The local spine now records the congruence, parallel, area, and square-construction tools that the proof uses.

Remark (Proof structure). Construct squares on the three sides of a right triangle, split the square on the hypotenuse into two rectangles, prove each rectangle equal to the corresponding square on a leg, and add the two equalities.

Remark (Dependencies).

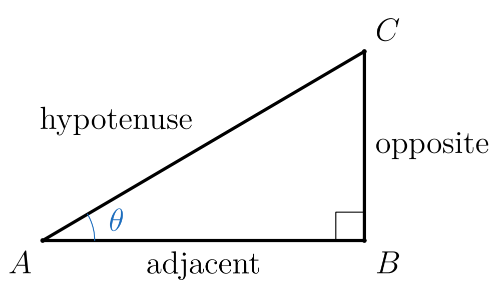
- Right angle and perpendicular
- Right-angled triangle
- Square
- Euclid Postulate I
- Euclid Common Notion II
- Euclid Common Notion III
- Euclid I.4
- Euclid I.41
- Euclid I.46

Chapter 2

Trigonometry

2.0.1 Trigonometric Functions from Euclidean Geometry

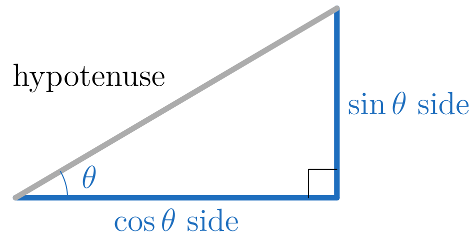
Remark (Geometric intuition). A right triangle fixes three side lengths relative to an acute angle: opposite, adjacent, and hypotenuse. Similar right triangles may be larger or smaller, but their corresponding side ratios are unchanged. This invariance is what makes a trigonometric ratio a function of the angle rather than a feature of one drawing.



Remark (Primitive and derived ratios). The primitive geometric ratios are sine and cosine. Tangent, cotangent, secant, and cosecant are convenient quotient or reciprocal ratios built from the same three side lengths.

Remark (Dependencies).

- Right-angled triangle
- Right angle and perpendicular



Definition

Definition 2.0.1 (Right-triangle sine)**.** Let T be a right triangle and let θ be one of its acute angles. Write

$$\mathrm{opp}(T,\theta), \qquad \mathrm{adj}(T,\theta), \qquad \mathrm{hyp}(T)$$

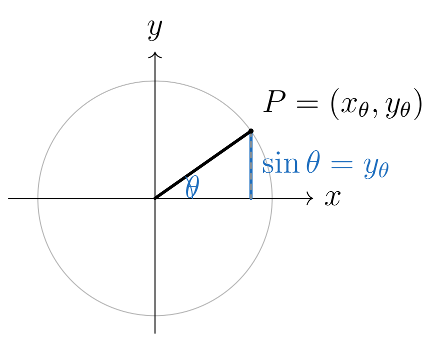
for the side lengths opposite θ, adjacent to θ, and opposite the right angle, respectively. Define

$$\sin\theta = \frac{\mathrm{opp}(T,\theta)}{\mathrm{hyp}(T)}.$$

Definition

Definition 2.0.2 (Acute-angle sine)**.** For an acute Euclidean angle θ, choose any right triangle T having θ as an acute angle and set

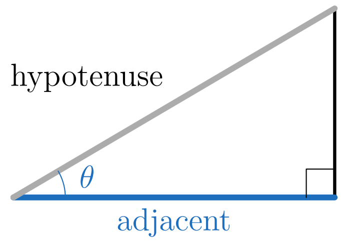
$$\sin\theta = \frac{\mathrm{opp}(T,\theta)}{\mathrm{hyp}(T)}.$$



Definition

Definition 2.0.3 (Unit-circle sine)**.** Let θ be an angle in standard position, and let $P(\theta) = (x_\theta, y_\theta)$ be the point where its terminal ray meets the unit circle. Define

$$\sin\theta = y_\theta.$$

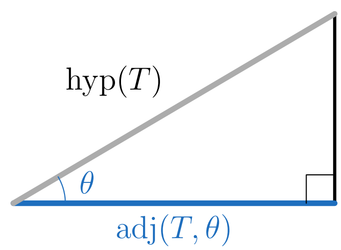


$\cos\theta = \text{adjacent}/\text{hypotenuse}$

Definition

Definition 2.0.4 (Right-triangle cosine)**.** Let T be a right triangle and let θ be one of its acute angles. Define

$$\cos\theta = \frac{\operatorname{adj}(T,\theta)}{\operatorname{hyp}(T)}.$$

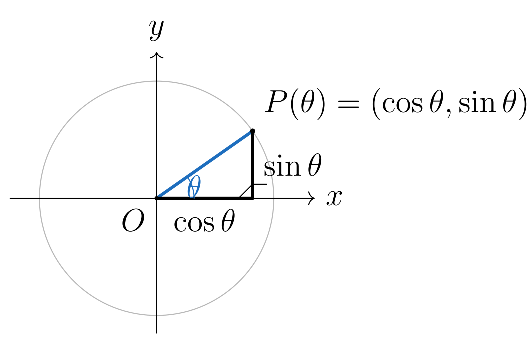


$\cos\theta = \operatorname{adj}(T,\theta)/\operatorname{hyp}(T)$

Definition

Definition 2.0.5 (Acute-angle cosine)**.** For an acute Euclidean angle θ, choose any right triangle T having θ as an acute angle and set

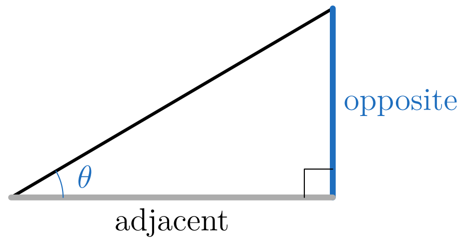
$$\cos\theta = \frac{\operatorname{adj}(T,\theta)}{\operatorname{hyp}(T)}.$$



Definition

Definition 2.0.6 (Unit-circle cosine)**.** Let θ be an angle in standard position, and let $P(\theta) = (x_\theta, y_\theta)$ be the point where its terminal ray meets the unit circle. Define

$$\cos\theta = x_\theta.$$

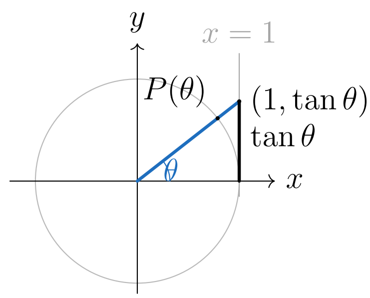


$\tan\theta = \text{opposite}/\text{adjacent}$

Definition

Definition 2.0.7 (Right-triangle tangent)**.** Let T be a right triangle and let θ be one of its acute angles. Define

$$\tan\theta = \frac{\operatorname{opp}(T,\theta)}{\operatorname{adj}(T,\theta)}.$$

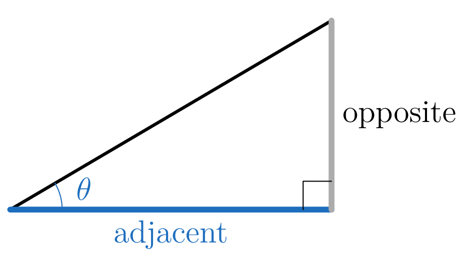


Definition

Definition 2.0.8 (Extended tangent)**.** Where $\cos\theta \neq 0$, define

$$\tan\theta = \frac{\sin\theta}{\cos\theta}.$$

Remark (Geometric reading). The tangent of θ is the signed height where the terminal ray meets the vertical tangent line $x = 1$.

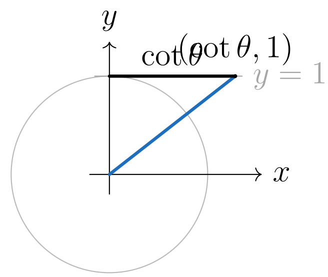


$\cot\theta = \text{adjacent}/\text{opposite}$

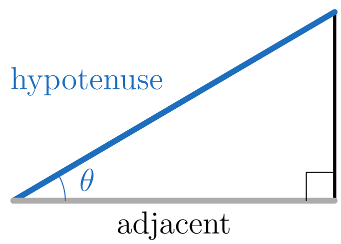
Definition

Definition 2.0.9 (Right-triangle cotangent)**.** Let T be a right triangle and let θ be one of its acute angles. Define

$$\cot\theta = \frac{\operatorname{adj}(T,\theta)}{\operatorname{opp}(T,\theta)}.$$

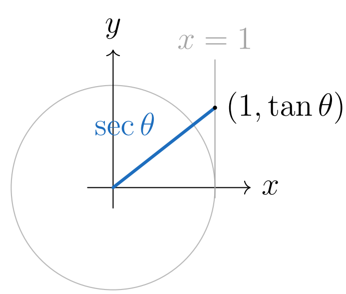


> **Definition**
>
> **Definition 2.0.10** (Extended cotangent)**.** Where $\sin\theta \neq 0$, define
>
> $$\cot\theta = \frac{\cos\theta}{\sin\theta}.$$

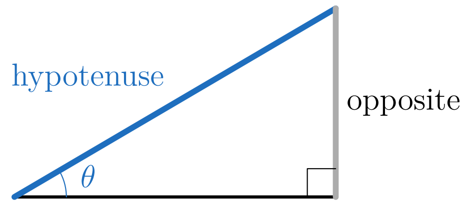


$\sec\theta = \text{hypotenuse}/\text{adjacent}$

> **Definition**
>
> **Definition 2.0.11** (Right-triangle secant)**.** Let T be a right triangle and let θ be one of its acute angles. Define
>
> $$\sec\theta = \frac{\text{hyp}(T)}{\text{adj}(T, \theta)}.$$



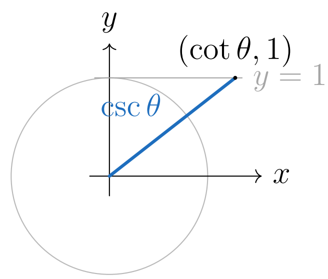
> **Definition**
>
> **Definition 2.0.12** (Extended secant)**.** Where $\cos\theta \neq 0$, define
>
> $$\sec\theta = \frac{1}{\cos\theta}.$$



$$\csc\theta = \text{hypotenuse}/\text{opposite}$$

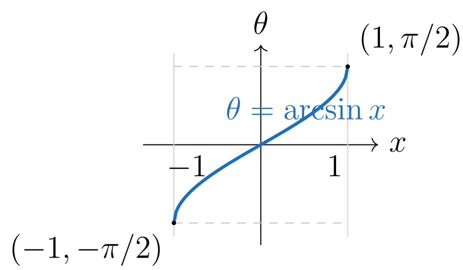
Definition

Definition 2.0.13 (Right-triangle cosecant)**.** Let T be a right triangle and let θ be one of its acute angles. Define
$$\csc\theta = \frac{\mathrm{hyp}(T)}{\mathrm{opp}(T,\theta)}.$$



Definition

Definition 2.0.14 (Extended cosecant)**.** Where $\sin\theta \neq 0$, define
$$\csc\theta = \frac{1}{\sin\theta}.$$



Definition

Definition 2.0.15 (Inverse sine)**.** The *inverse sine* function is the inverse of sine after sine is restricted to the interval
$$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right].$$
It is denoted by arcsin or $\sin^{-1}$ and has domain $[-1, 1]$ and range $[-\pi/2, \pi/2]$.
$$\arcsin x = \theta \quad\Longleftrightarrow\quad \sin\theta = x \quad\text{and}\quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}.$$

Remark (Domain and range)*.* The full sine function has domain all real-valued angle measures and range $[-1, 1]$. The inverse sine function reverses the restricted sine function, so its domain is $[-1, 1]$ and its range is the principal interval $[-\pi/2, \pi/2]$.

Remark (Dependencies)*.*

- Unit-circle sine
- Radian measure

Theorem

Theorem 2.0.16 (Principal inverse sine identities)**.** *For every* $x \in [-1, 1]$*,*

$$\sin(\arcsin x) = x.$$

For every $\theta \in [-\pi/2, \pi/2]$*,*

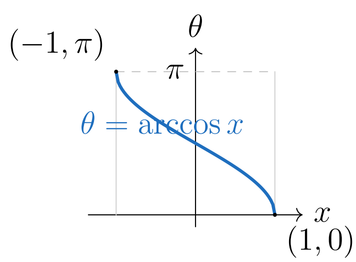
$$\arcsin(\sin\theta) = \theta.$$

Remark (Standard quantified statement)*.*

$$\forall x \in [-1, 1], \quad \sin(\arcsin x) = x, \qquad \forall\theta \in [-\pi/2, \pi/2], \quad \arcsin(\sin\theta) = \theta.$$

Remark (Dependencies)*.*

- Inverse sine



Definition

Definition 2.0.17 (Inverse cosine)**.** The *inverse cosine* function is the inverse of cosine after cosine is restricted to the interval $[0, \pi]$. It is denoted by arccos or $\cos^{-1}$ and has domain $[-1, 1]$ and range $[0, \pi]$.

$$\arccos x = \theta \quad \iff \quad \cos\theta = x \quad \text{and} \quad 0 \le \theta \le \pi.$$

Remark (Domain and range)*.* The full cosine function has domain all real-valued angle measures and range $[-1, 1]$. The inverse cosine function reverses the restricted cosine function, so its domain is $[-1, 1]$ and its range is the principal interval $[0, \pi]$.

Remark (Dependencies)*.*

- Unit-circle cosine

- Radian measure

Theorem

Theorem 2.0.18 (Principal inverse cosine identities)**.** *For every $x \in [-1, 1]$,*

$$\cos(\arccos x) = x.$$

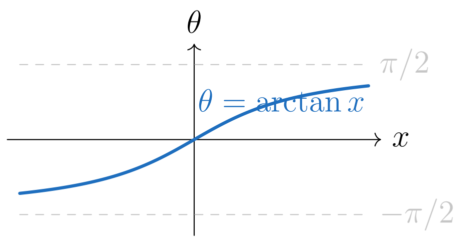
For every $\theta \in [0, \pi]$,

$$\arccos(\cos \theta) = \theta.$$

Remark (Standard quantified statement)*.* $\forall x \in [-1, 1]$, $\cos(\arccos x) = x$, and $\forall \theta \in [0, \pi]$, $\arccos(\cos \theta) = \theta$.

Remark (Dependencies)*.*

- Inverse cosine



Definition

Definition 2.0.19 (Inverse tangent)**.** The *inverse tangent* function is the inverse of tangent after tangent is restricted to the interval $(-\pi/2, \pi/2)$. It is denoted by arctan or $\tan^{-1}$ and has domain $\mathbb{R}$ and range $(-\pi/2, \pi/2)$.

$$\arctan x = \theta \quad \iff \quad \tan \theta = x \quad \text{and} \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}.$$

Remark (Domain and range)*.* The tangent function has domain $\mathbb{R} \setminus \{\pi/2 + k\pi : k \in \mathbb{Z}\}$ and range $\mathbb{R}$. The inverse tangent function reverses the principal restricted tangent function, so its domain is $\mathbb{R}$ and its range is $(-\pi/2, \pi/2)$.

Remark (Dependencies)*.*

- Extended tangent
- Radian measure

Theorem

Theorem 2.0.20 (Principal inverse tangent identities)**.** *For every $x \in \mathbb{R}$,*

$$\tan(\arctan x) = x.$$

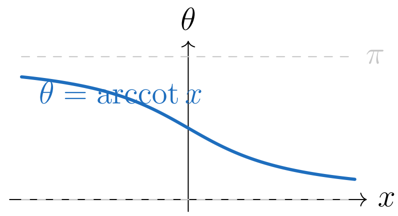
For every $\theta \in (-\pi/2, \pi/2)$,

$$\arctan(\tan \theta) = \theta.$$

Remark (Standard quantified statement). $\forall x \in \mathbb{R}$, $\tan(\arctan x) = x$, and $\forall \theta \in (-\pi/2, \pi/2)$, $\arctan(\tan \theta) = \theta$.

Remark (Dependencies).

- Inverse tangent



Definition

Definition 2.0.21 (Inverse cotangent). The *inverse cotangent* function is the inverse of cotangent after cotangent is restricted to the interval $(0, \pi)$. It is denoted by arccot or $\cot^{-1}$ and has domain $\mathbb{R}$ and range $(0, \pi)$.

$$\operatorname{arccot} x = \theta \quad \Longleftrightarrow \quad \cot \theta = x \quad \text{and} \quad 0 < \theta < \pi.$$

Remark (Domain and range). The cotangent function has domain $\mathbb{R} \setminus \{k\pi : k \in \mathbb{Z}\}$ and range $\mathbb{R}$. The inverse cotangent function reverses the principal restricted cotangent function, so its domain is $\mathbb{R}$ and its range is $(0, \pi)$.

Remark (Dependencies).

- Extended cotangent
- Radian measure

Theorem

Theorem 2.0.22 (Principal inverse cotangent identities). *For every $x \in \mathbb{R}$,*

$$\cot(\operatorname{arccot} x) = x.$$

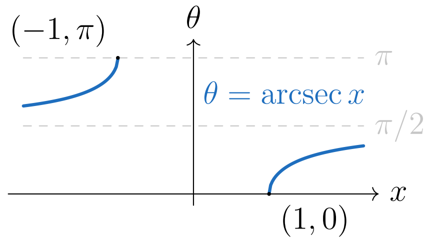
For every $\theta \in (0, \pi)$,

$$\operatorname{arccot}(\cot \theta) = \theta.$$

Remark (Standard quantified statement). $\forall x \in \mathbb{R}$, $\cot(\operatorname{arccot} x) = x$, and $\forall \theta \in (0, \pi)$, $\operatorname{arccot}(\cot \theta) = \theta$.

Remark (Dependencies).

- Inverse cotangent



Definition

Definition 2.0.23 (Inverse secant)**.** The *inverse secant* function is the inverse of secant after secant is restricted to $[0, \pi] \setminus \{\pi/2\}$. It is denoted by arcsec or $\sec^{-1}$ and has domain $(-\infty, -1] \cup [1, \infty)$ and range $[0, \pi] \setminus \{\pi/2\}$.

$$\operatorname{arcsec} x = \theta \iff \sec\theta = x \quad \text{and} \quad 0 \leq \theta \leq \pi,\ \theta \neq \frac{\pi}{2}.$$

Remark (Domain and range). The secant function has domain $\mathbb{R} \setminus \{\pi/2 + k\pi : k \in \mathbb{Z}\}$ and range $(-\infty, -1] \cup [1, \infty)$. The inverse secant function reverses the principal restricted secant function.

Remark (Dependencies).

- Extended secant
- Radian measure

Theorem

Theorem 2.0.24 (Principal inverse secant identities)**.** *For every $x \in (-\infty, -1] \cup [1, \infty)$,*

$$\sec(\operatorname{arcsec} x) = x.$$

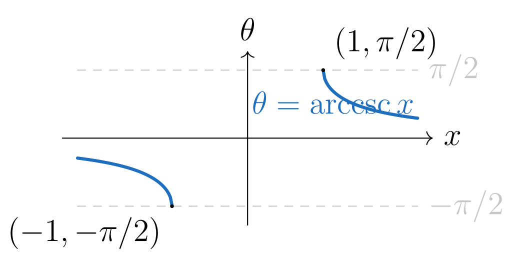
For every $\theta \in [0, \pi] \setminus \{\pi/2\}$,

$$\operatorname{arcsec}(\sec\theta) = \theta.$$

Remark (Standard quantified statement). $\forall x \in (-\infty, -1] \cup [1, \infty)$, $\sec(\operatorname{arcsec} x) = x$, and $\forall \theta \in [0, \pi] \setminus \{\pi/2\}$, $\operatorname{arcsec}(\sec\theta) = \theta$.

Remark (Dependencies).

- Inverse secant



Definition

Definition 2.0.25 (Inverse cosecant)**.** The *inverse cosecant* function is the inverse of cosecant after cosecant is restricted to $[-\pi/2, \pi/2] \setminus \{0\}$. It is denoted by arccsc or $\csc^{-1}$ and has domain $(-\infty, -1] \cup [1, \infty)$ and range $[-\pi/2, \pi/2] \setminus \{0\}$.

$$\operatorname{arccsc} x = \theta \iff \csc\theta = x \quad \text{and} \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2},\ \theta \neq 0.$$

Remark (Domain and range). The cosecant function has domain $\mathbb{R} \setminus \{k\pi : k \in \mathbb{Z}\}$ and range $(-\infty, -1] \cup [1, \infty)$. The inverse cosecant function reverses the principal restricted cosecant function.

Remark (Dependencies).

- Extended cosecant
- Radian measure

Theorem

Theorem 2.0.26 (Principal inverse cosecant identities)**.** *For every* $x \in (-\infty, -1] \cup [1, \infty)$,

$$\csc(\operatorname{arccsc} x) = x.$$

For every $\theta \in [-\pi/2, \pi/2] \setminus \{0\}$,

$$\operatorname{arccsc}(\csc\theta) = \theta.$$

Remark (Standard quantified statement). $\forall x \in (-\infty, -1] \cup [1, \infty)$, $\csc(\operatorname{arccsc} x) = x$, and $\forall\theta \in [-\pi/2, \pi/2] \setminus \{0\}$, $\operatorname{arccsc}(\csc\theta) = \theta$.

Remark (Dependencies).

- Inverse cosecant

Theorem

Theorem 2.0.27 (Cofunction identities for inverse sine and inverse cosine)**.** *For every* $x \in [-1, 1]$,

$$\arcsin x + \arccos x = \frac{\pi}{2}.$$

Equivalently,

$$\arccos x = \frac{\pi}{2} - \arcsin x, \qquad \arcsin x = \frac{\pi}{2} - \arccos x.$$

Remark (Standard quantified statement). $\forall x \in [-1, 1]$, $\arcsin x + \arccos x = \pi/2$.

Remark (Dependencies).

- Inverse sine

- Inverse cosine

Theorem

Theorem 2.0.28 (Reciprocal identities for inverse secant and inverse cosecant)**.** *For every $x \in (-\infty, -1] \cup [1, \infty)$,*

$$\operatorname{arcsec} x = \arccos\left(\frac{1}{x}\right), \qquad \operatorname{arccsc} x = \arcsin\left(\frac{1}{x}\right).$$

Remark (Standard quantified statement)*.*

$$\forall x \in (-\infty, -1] \cup [1, \infty), \quad \operatorname{arcsec} x = \arccos(1/x) \quad \text{and} \quad \operatorname{arccsc} x = \arcsin(1/x).$$

Remark (Dependencies)*.*

- Inverse cosine
- Inverse sine
- Inverse secant
- Inverse cosecant

Theorem

Theorem 2.0.29 (Reciprocal identity for inverse cotangent)**.** *For every nonzero real number x,*

$$\operatorname{arccot} x = \begin{cases} \arctan(1/x), & x > 0, \\ \pi + \arctan(1/x), & x < 0. \end{cases}$$

Remark (Branch convention)*.* The second case is necessary because this chapter uses the principal range $(0, \pi)$ for arccot and $(-\pi/2, \pi/2)$ for arctan.

Remark (Dependencies)*.*

- Inverse tangent
- Inverse cotangent

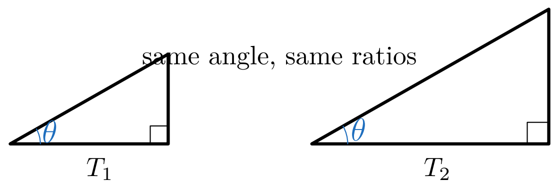
Remark (Standard quantified statement)*.* For every right triangle T and every acute angle θ of T, the six displayed ratios are defined whenever the denominator in the displayed quotient is nonzero.

Remark (Standard quantified statement)*.* For every acute Euclidean angle θ, if T is a right triangle having θ as an acute angle, then $\sin\theta$ and $\cos\theta$ are the opposite-over-hypotenuse and adjacent-over-hypotenuse ratios of T.

Remark (Well-definedness debt)*.* These definitions depend on the similarity-invariance theorem below. Until that theorem is invoked, the displayed formulas describe a ratio in a chosen triangle, not yet a function of θ alone.

Remark (Dependencies).

- Right-triangle sine
- Right-triangle cosine



Theorem

Theorem 2.0.30 (Similarity invariance of trigonometric ratios)**.** *If two right triangles have the same acute angle θ, then their corresponding opposite-to-hypotenuse and adjacent-to-hypotenuse ratios are equal.*

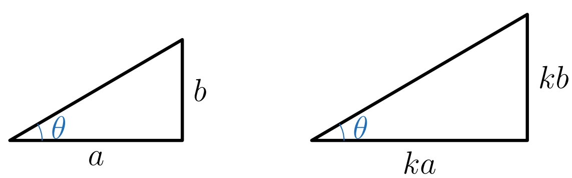
Remark (Proof)*.* Go to proof.

Remark (Standard quantified statement)*.* For all right triangles T_1, T_2 and every acute angle θ, if θ is an acute angle of both triangles, then

$$\frac{\mathrm{opp}(T_1,\theta)}{\mathrm{hyp}(T_1)} = \frac{\mathrm{opp}(T_2,\theta)}{\mathrm{hyp}(T_2)} \quad \text{and} \quad \frac{\mathrm{adj}(T_1,\theta)}{\mathrm{hyp}(T_1)} = \frac{\mathrm{adj}(T_2,\theta)}{\mathrm{hyp}(T_2)}.$$

Remark (Dependencies).

- Euclid I.32



same angle, same ratios

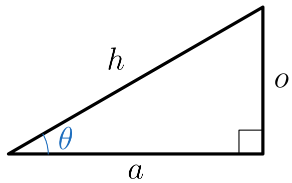
Theorem

Theorem 2.0.31 (Well-definedness of sine and cosine)**.** *For each acute Euclidean angle θ, the values $\sin\theta$ and $\cos\theta$ are independent of the right triangle chosen to compute them.*

Remark (Proof)*.* Go to proof.

Remark (Dependencies).

- Acute-angle sine
- Acute-angle cosine
- Similarity invariance of trigonometric ratios



$$\sin = o/h, \quad \cos = a/h$$
$$\tan = o/a, \quad \cot = a/o$$
$$\sec = h/a, \quad \csc = h/o$$

Theorem

Theorem 2.0.32 (Six trigonometric functions)**.** *The six right-triangle trigonometric ratios define functions of an acute Euclidean angle, subject to the denominator restrictions in their defining quotients.*

Remark (Proof)*.* Go to proof.

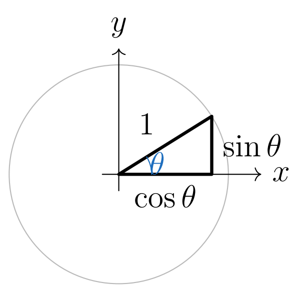
Remark (Dependencies)*.*

- Right-triangle sine
- Right-triangle cosine
- Well-definedness of sine and cosine

Remark (Debt flag)*.* This definition extends sine and cosine beyond acute angles. Treating θ as an arbitrary real-valued radian input requires arc length and rectifiability machinery not yet developed in this volume.

Remark (Dependencies)*.*

- Circle
- Cartesian Product
- Well-definedness of sine and cosine



Theorem

Theorem 2.0.33 (Unit-circle agreement with right-triangle trigonometry)**.** *For* $0 < \theta < 90°$, *the unit-circle definitions of sine and cosine agree with the right-triangle definitions.*

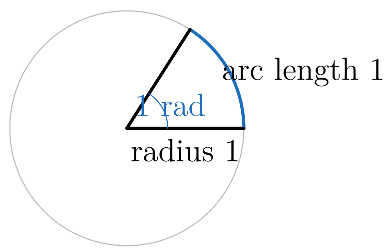
Remark (Proof)*.* Go to proof.

Remark (Dependencies)*.*

- Acute-angle sine
- Acute-angle cosine
- Unit-circle cosine
- Unit-circle sine

Remark (Dependencies).

- Unit-circle cosine
- Unit-circle sine
- Unit-circle agreement with right-triangle trigonometry



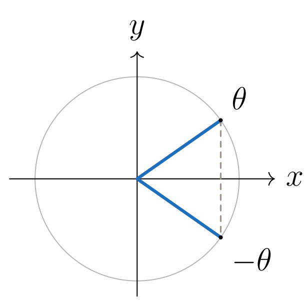
Definition

Definition 2.0.34 (Radian measure)**.** One radian is the angle that cuts off arc length 1 on the unit circle. A full turn has measure 2π radians.

Remark (Debt flag). This definition uses arc length intuitively. Its rigorous foundation belongs after rectifiable curves and the construction of π from circumference.

Remark (Dependencies).

- Unit-circle cosine
- Unit-circle sine
- Circle



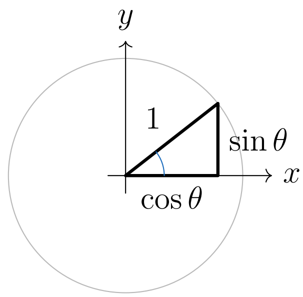
Theorem

Theorem 2.0.35 (Symmetry and periodicity of sine and cosine). *For every angle in the unit-circle domain,*

$$\cos(-\theta) = \cos\theta, \qquad \sin(-\theta) = -\sin\theta,$$

and one full turn returns the same sine and cosine values.

Remark (Proof). Go to proof.



Theorem

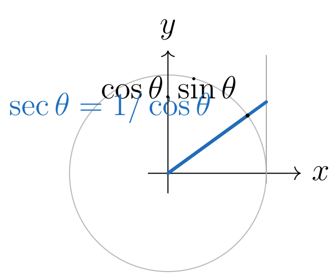
Theorem 2.0.36 (Pythagorean identity). *For every angle in the unit-circle domain,*

$$\cos^2\theta + \sin^2\theta = 1.$$

Remark (Proof). Go to proof.

Remark (Dependencies).

- Euclid I.47
- Unit-circle cosine
- Unit-circle sine



Theorem

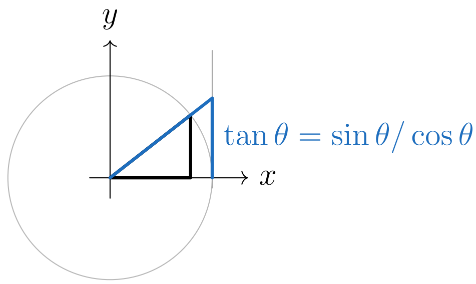
Theorem 2.0.37 (Reciprocal identities). *Where both sides are defined,*

$$\sec\theta = \frac{1}{\cos\theta}, \qquad \csc\theta = \frac{1}{\sin\theta}, \qquad \cot\theta = \frac{1}{\tan\theta}.$$

Remark (Proof). Go to proof.

Remark (Dependencies).

- Extended tangent
- Extended cotangent
- Extended secant
- Extended cosecant



Theorem

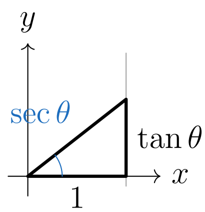
Theorem 2.0.38 (Quotient identities)**.** *Where both sides are defined,*

$$\tan\theta = \frac{\sin\theta}{\cos\theta}, \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}.$$

Remark (Proof). Go to proof.

Remark (Dependencies).

- Extended tangent
- Extended cotangent



Theorem

Theorem 2.0.39 (Tangent and secant Pythagorean identities)**.** *Where both sides are defined,*

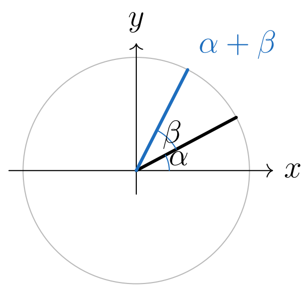
$$1 + \tan^2\theta = \sec^2\theta, \qquad 1 + \cot^2\theta = \csc^2\theta.$$

Remark (Proof). Go to proof.

Remark (Dependencies).

- Pythagorean identity

- Quotient identities
- Reciprocal identities



Theorem

Theorem 2.0.40 (Angle-sum identities)**.** *For angles in the unit-circle domain,*

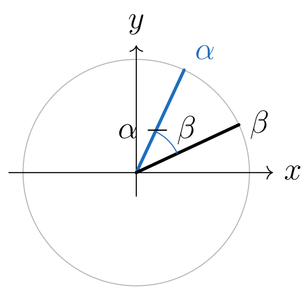
$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta,$$

$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta.$$

Remark (Proof)*.* Go to proof.

Remark (Dependencies)*.*

- Pythagorean identity



Theorem

Theorem 2.0.41 (Angle-difference identities)**.** *For angles in the unit-circle domain,*

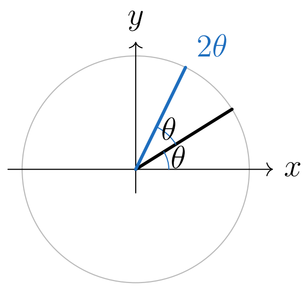
$$\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta,$$

$$\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta.$$

Remark (Proof)*.* Go to proof.

Remark (Dependencies)*.*

- Angle-sum identities
- Symmetry and periodicity of sine and cosine



Theorem

Theorem 2.0.42 (Double-angle identities)**.** *For angles in the unit-circle domain,*

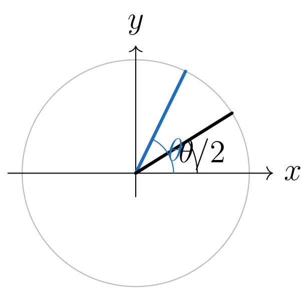
$$\sin(2\theta) = 2\sin\theta\cos\theta,$$

$$\cos(2\theta) = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta.$$

Remark (Proof)*.* Go to proof.

Remark (Dependencies)*.*

- Angle-sum identities
- Pythagorean identity



Theorem

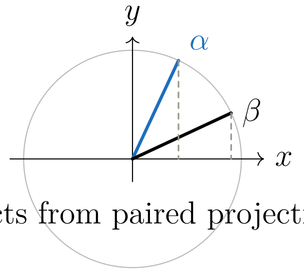
Theorem 2.0.43 (Half-angle identities)**.** *Where the signs are chosen according to the quadrant of $\theta/2$,*

$$\sin\frac{\theta}{2} = \pm\sqrt{\frac{1-\cos\theta}{2}}, \qquad \cos\frac{\theta}{2} = \pm\sqrt{\frac{1+\cos\theta}{2}}.$$

Remark (Proof)*.* Go to proof.

Remark (Dependencies)*.*

- Double-angle identities



products from paired projections

Theorem

Theorem 2.0.44 (Product-to-sum identities)**.** *For angles in the unit-circle domain,*

$$\sin\alpha\cos\beta = \frac{\sin(\alpha+\beta)+\sin(\alpha-\beta)}{2},$$

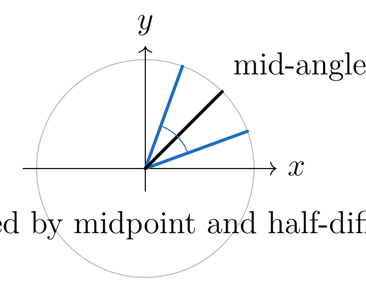
$$\cos\alpha\cos\beta = \frac{\cos(\alpha+\beta)+\cos(\alpha-\beta)}{2},$$

$$\sin\alpha\sin\beta = \frac{\cos(\alpha-\beta)-\cos(\alpha+\beta)}{2}.$$

Remark (Proof)*.* Go to proof.

Remark (Dependencies).

- Angle-sum identities
- Angle-difference identities



sums resolved by midpoint and half-difference

Theorem

Theorem 2.0.45 (Sum-to-product identities)**.** *For angles in the unit-circle domain,*

$$\sin u + \sin v = 2\sin\frac{u+v}{2}\cos\frac{u-v}{2},$$

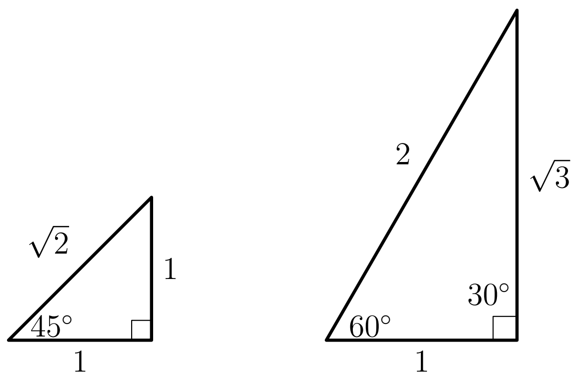
$$\cos u + \cos v = 2\cos\frac{u+v}{2}\cos\frac{u-v}{2},$$

$$\cos u - \cos v = -2\sin\frac{u+v}{2}\sin\frac{u-v}{2}.$$

Remark (Proof)*.* Go to proof.

Remark (Dependencies).

- Product-to-sum identities



Remark (Special angles). The 45° and 30°–60° values come from the isosceles right triangle and the bisected equilateral triangle. The radian entries use the radian convention above and inherit its arc-length debt.

2.0.2 Hyperbolic Trigonometric Functions from Geometry

Remark (Motivating picture). Circular trigonometry reads coordinates from the unit circle $x^2 + y^2 = 1$. Hyperbolic trigonometry uses the same geometric idea with the unit rectangular hyperbola

$$x^2 - y^2 = 1$$

in place of the circle. The parameter is not an angle. It is twice the signed area of the hyperbolic sector swept from the vertex $(1, 0)$ to the point on the right branch.

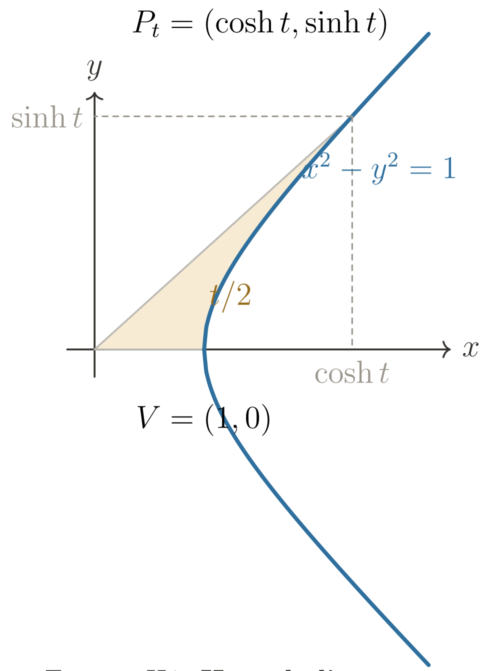


FIGURE H1. **Hyperbolic sector:**
The parameter t is twice the shaded sector area.

Remark (Construction versus behavior). This chapter gives the geometric construction. Continuity and differentiation statements are recorded below as forward pointers to the later analysis chapter on the behavior of elementary functions, where limits, continuity, and differentiation provide the canonical proofs.

Remark (Dependencies).

- Cartesian Product
- Radian measure

Definition

Definition 2.0.46 (Hyperbolic sector)**.** Let $O = (0, 0)$ and $V = (1, 0)$. For a point $P = (u, v)$ on the right branch of $x^2 - y^2 = 1$, write

$$\mathrm{HArea}(O, V, P)$$

for the signed area of the region bounded by the segment OV, the segment OP, and the hyperbolic arc from V to P. The sign is positive on the upper branch and negative on the lower branch.

Remark (Predicate form).

$$\mathrm{Sector}(P, t) \iff P = (u, v),\ u^2 - v^2 = 1,\ u > 0,\ \text{and} 2\,\mathrm{HArea}(O, V, P) = t.$$

Remark (Dependencies).

- Cartesian Product

Definition

Definition 2.0.47 (Swept hyperbola point)**.** For $t \geq 0$, let P_t be the unique point on the upper right branch of $x^2 - y^2 = 1$ satisfying

$$2\,\mathrm{HArea}(O, V, P_t) = t.$$

For $t < 0$, let P_t be the reflection of P_{-t} across the x-axis.

Remark (Coordinate notation). The separate hyperbolic sine and cosine definitions below name the two coordinates of this same point. They are not independent constructions.

Remark (Dependencies).

- Hyperbolic sector

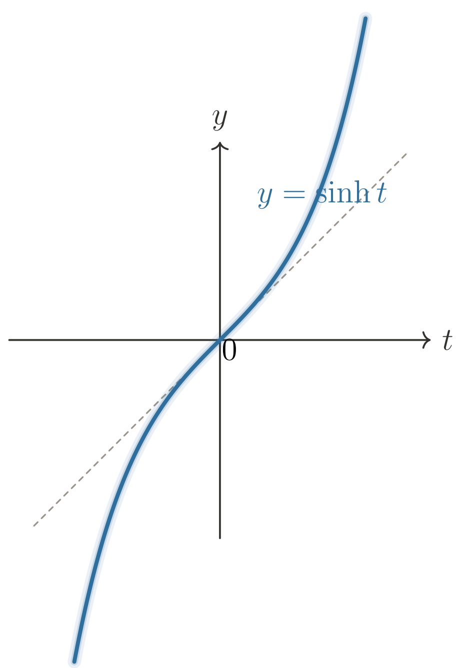


Figure H2. **Hyperbolic sine:**
Odd, strictly increasing, and unbounded in both directions.

Definition

Definition 2.0.48 (Hyperbolic sine)**.** For each $t \in \mathbb{R}$, define $\sinh t$ to be the second coordinate of the swept hyperbola point P_t:

$$P_t = (u, \sinh t).$$

Remark (Domain and range). The function $\sinh : \mathbb{R} \to \mathbb{R}$ has domain all real numbers and range $\mathbb{R}$. It is odd by reflection of the swept point across the x-axis.

Remark (Dependencies).

- Swept hyperbola point

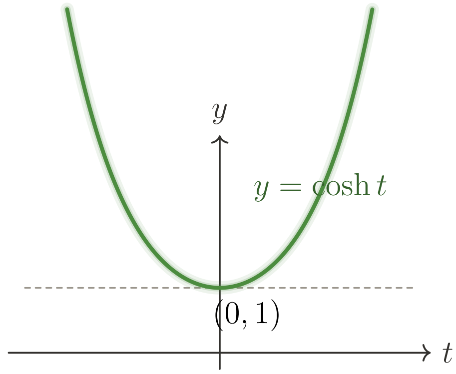


Figure H3. **Hyperbolic cosine: Even, with global minimum 1 at $t = 0$.**

Definition

Definition 2.0.49 (Hyperbolic cosine)**.** For each $t \in \mathbb{R}$, define $\cosh t$ to be the first coordinate of the swept hyperbola point P_t:

$$P_t = (\cosh t, v).$$

Remark (Domain and range)*.* The function $\cosh : \mathbb{R} \to \mathbb{R}$ has domain all real numbers and range $[1, \infty)$. It is even by reflection of the swept point across the x-axis.

Remark (Dependencies).

- Swept hyperbola point

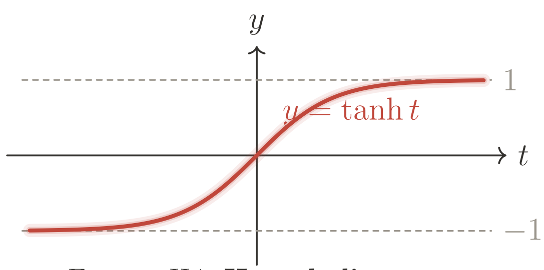


Figure H4. **Hyperbolic tangent: Odd, increasing, with horizontal asymptotes $y = \pm 1$.**

Definition

Definition 2.0.50 (Hyperbolic tangent)**.** For each $t \in \mathbb{R}$, define the *hyperbolic tangent* by

$$\tanh t = \frac{\sinh t}{\cosh t}.$$

Remark (Domain and range)*.* The function tanh is defined on all of $\mathbb{R}$ because $\cosh t \geq 1$. Its range is $(-1, 1)$.

Remark (Dependencies).

- Hyperbolic sine
- Hyperbolic cosine

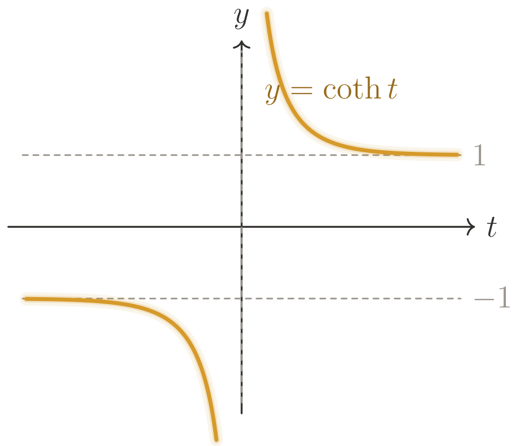


FIGURE H5. **Hyperbolic cotangent:**
Undefined at $t = 0$, with asymptotes $y = \pm 1$.

Definition

Definition 2.0.51 (Hyperbolic cotangent)**.** For each $t \in \mathbb{R} \setminus \{0\}$, define the *hyperbolic cotangent* by

$$\coth t = \frac{\cosh t}{\sinh t}.$$

Remark (Domain and range). The function coth is defined on $\mathbb{R} \setminus \{0\}$ because $\sinh 0 = 0$. Its range is $(-\infty, -1) \cup (1, \infty)$.

Remark (Dependencies).

- Hyperbolic sine
- Hyperbolic cosine

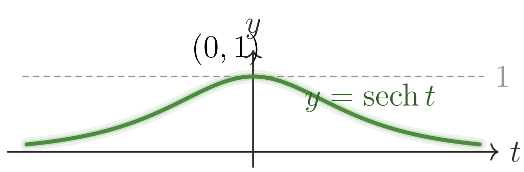


FIGURE H6. **Hyperbolic secant:**
Even, bell-shaped, with maximum 1 at $t = 0$.

Definition

Definition 2.0.52 (Hyperbolic secant)**.** For each $t \in \mathbb{R}$, define the *hyperbolic secant* by

$$\operatorname{sech} t = \frac{1}{\cosh t}.$$

Remark (Domain and range). The function sech is defined on all of $\mathbb{R}$ because $\cosh t \geq 1$. Its range is $(0, 1]$.

Remark (Dependencies).

- Hyperbolic cosine

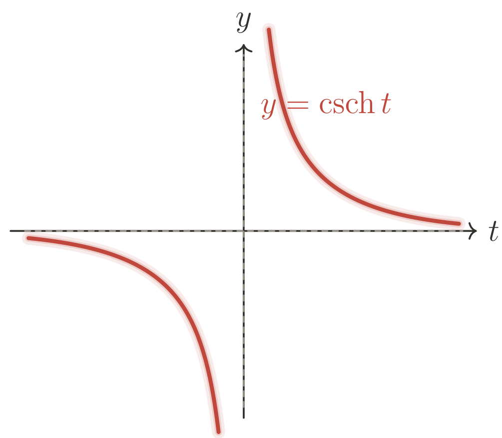


Figure H7. **Hyperbolic cosecant:**
Odd, undefined at $t = 0$, and decays toward 0.

Definition

Definition 2.0.53 (Hyperbolic cosecant)**.** For each $t \in \mathbb{R} \setminus \{0\}$, define the *hyperbolic cosecant* by

$$\operatorname{csch} t = \frac{1}{\sinh t}.$$

Remark (Domain and range). The function csch is defined on $\mathbb{R} \setminus \{0\}$ because $\sinh 0 = 0$. Its range is $\mathbb{R} \setminus \{0\}$.

Remark (Dependencies).

- Hyperbolic sine

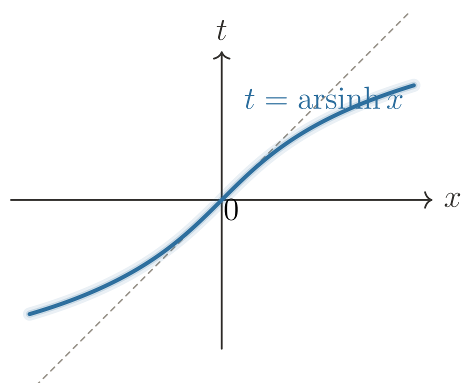


Figure HI1. **Inverse hyperbolic sine:**
Domain and range $\mathbb{R}$; the graph reflects $y = \sinh t$ across $y = x$.

Definition

Definition 2.0.54 (Inverse hyperbolic sine)**.** The *inverse hyperbolic sine* function is the inverse of sinh on its full domain. It is denoted by arsinh and has domain $\mathbb{R}$ and range $\mathbb{R}$.

$$\operatorname{arsinh} x = t \quad \Longleftrightarrow \quad \sinh t = x.$$

Remark (Domain and range). The regular function $\sinh : \mathbb{R} \to \mathbb{R}$ is strictly increasing and has range $\mathbb{R}$. No branch restriction is needed before inverting it.

Remark (Dependencies).

- Hyperbolic sine
- Existence for hyperbolic sectors
- Uniqueness for hyperbolic sectors

Theorem

Theorem 2.0.55 (Principal inverse hyperbolic sine identities)**.** *For every $x \in \mathbb{R}$,*

$$\sinh(\operatorname{arsinh} x) = x.$$

For every $t \in \mathbb{R}$,

$$\operatorname{arsinh}(\sinh t) = t.$$

Remark (Proof). Both identities are the defining cancellation laws for inverse functions, using that sinh is one-to-one from $\mathbb{R}$ onto $\mathbb{R}$.

Remark (Dependencies).

- Inverse hyperbolic sine

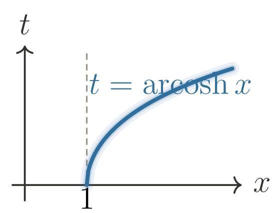


FIGURE HI2. **Inverse hyperbolic cosine:**
Domain $[1, \infty)$ and range $[0, \infty)$; the inverse uses the nonnegative branch.

Definition

Definition 2.0.56 (Inverse hyperbolic cosine)**.** The *inverse hyperbolic cosine* function is the inverse of cosh after cosh is restricted to the interval $[0, \infty)$. It is denoted by arcosh and has domain $[1, \infty)$ and range $[0, \infty)$.

$$\operatorname{arcosh} x = t \quad \Longleftrightarrow \quad \cosh t = x \quad \text{and} \quad t \geq 0.$$

Remark (Domain and range). The regular function $\cosh : \mathbb{R} \to [1, \infty)$ is even, so it is not one-to-one on all of $\mathbb{R}$. The principal inverse uses the nonnegative branch, where cosh increases from 1 to infinity.

Remark (Dependencies).

- Hyperbolic cosine
- Existence for hyperbolic sectors
- Uniqueness for hyperbolic sectors

Theorem

Theorem 2.0.57 (Principal inverse hyperbolic cosine identities)**.** *For every $x \in [1, \infty)$,*

$$\cosh(\operatorname{arcosh} x) = x.$$

For every $t \in [0, \infty)$,

$$\operatorname{arcosh}(\cosh t) = t.$$

Remark (Proof). These are the cancellation laws for the inverse of the restricted function $\cosh|_{[0,\infty)} : [0, \infty) \to [1, \infty)$.

Remark (Dependencies).

- Inverse hyperbolic cosine

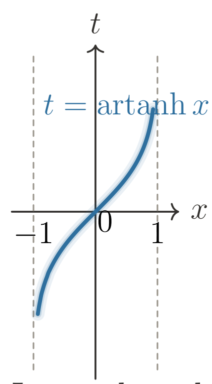


FIGURE HI3. **Inverse hyperbolic tangent:**
Domain $(-1, 1)$ and range $\mathbb{R}$; the vertical lines $x = \pm 1$ are asymptotes.

Definition

Definition 2.0.58 (Inverse hyperbolic tangent)**.** The *inverse hyperbolic tangent* function is the inverse of tanh on its full domain. It is denoted by artanh and has domain $(-1, 1)$ and range $\mathbb{R}$.

$$\operatorname{artanh} x = t \iff \tanh t = x.$$

Remark (Domain and range). The regular function $\tanh : \mathbb{R} \to (-1, 1)$ is strictly increasing. Its horizontal asymptotes show that the endpoint values -1 and 1 are not attained, so they are not in the inverse domain.

Remark (Dependencies).

- Hyperbolic tangent

Theorem

Theorem 2.0.59 (Principal inverse hyperbolic tangent identities)**.** *For every $x \in (-1, 1)$,*

$$\tanh(\operatorname{artanh} x) = x.$$

For every $t \in \mathbb{R}$,

$$\operatorname{artanh}(\tanh t) = t.$$

Remark (Proof)*.* Both identities are the cancellation laws for the inverse of $\tanh : \mathbb{R} \to (-1, 1)$.

Remark (Dependencies).

- Inverse hyperbolic tangent

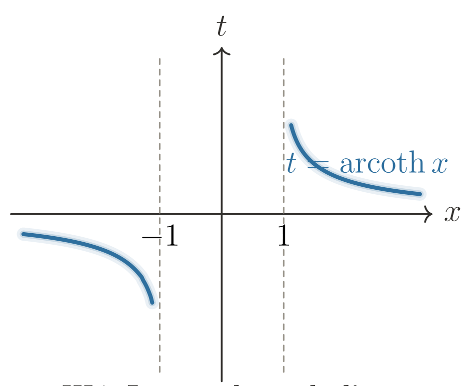


Figure HI4. **Inverse hyperbolic cotangent:**
Domain $(-\infty, -1) \cup (1, \infty)$ and range $\mathbb{R} \setminus \{0\}$; both branches are retained.

Definition

Definition 2.0.60 (Inverse hyperbolic cotangent)**.** The *inverse hyperbolic cotangent* function is the inverse of coth on its two natural branches. It is denoted by arcoth and has domain $(-\infty, -1) \cup (1, \infty)$ and range $\mathbb{R} \setminus \{0\}$.

$$\operatorname{arcoth} x = t \quad \iff \quad \coth t = x \quad \text{and} \quad t \neq 0.$$

Remark (Domain and range)*.* The regular function coth is undefined at 0. On $(0, \infty)$ its range is $(1, \infty)$, and on $(-\infty, 0)$ its range is $(-\infty, -1)$. The inverse keeps both branches and excludes 0 from its range.

Remark (Dependencies).

- Hyperbolic cotangent

Theorem

Theorem 2.0.61 (Principal inverse hyperbolic cotangent identities)**.** *For every* $x \in (-\infty, -1) \cup (1, \infty)$,

$$\coth(\operatorname{arcoth} x) = x.$$

For every $t \in \mathbb{R} \setminus \{0\}$,

$$\operatorname{arcoth}(\coth t) = t.$$

Remark (Proof). The positive and negative branches of coth are each one-to-one, and their ranges are disjoint. Therefore the two-branch inverse has the displayed cancellation laws.

Remark (Dependencies).

- Inverse hyperbolic cotangent

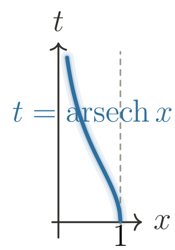


FIGURE HI5. **Inverse hyperbolic secant:**
Domain $(0, 1]$ **and range** $[0, \infty)$**; the inverse uses the nonnegative branch of** sech**.**

Definition

Definition 2.0.62 (Inverse hyperbolic secant)**.** The *inverse hyperbolic secant* function is the inverse of sech after sech is restricted to $[0, \infty)$. It is denoted by arsech and has domain $(0, 1]$ and range $[0, \infty)$.

$$\operatorname{arsech} x = t \iff \operatorname{sech} t = x \quad \text{and} \quad t \geq 0.$$

Remark (Domain and range). The regular function $\operatorname{sech} : \mathbb{R} \to (0, 1]$ is even, so it is not one-to-one on all of $\mathbb{R}$. The principal inverse uses the nonnegative branch, where sech decreases from 1 toward 0 without attaining 0.

Remark (Dependencies).

- Hyperbolic secant

Theorem

Theorem 2.0.63 (Principal inverse hyperbolic secant identities)**.** *For every* $x \in (0, 1]$,

$$\operatorname{sech}(\operatorname{arsech} x) = x.$$

For every $t \in [0, \infty)$,

$$\operatorname{arsech}(\operatorname{sech} t) = t.$$

Remark (Proof). These are the cancellation laws for the inverse of the restricted function $\operatorname{sech}|_{[0,\infty)} : [0, \infty) \to (0, 1]$.

Remark (Dependencies).

- Inverse hyperbolic secant

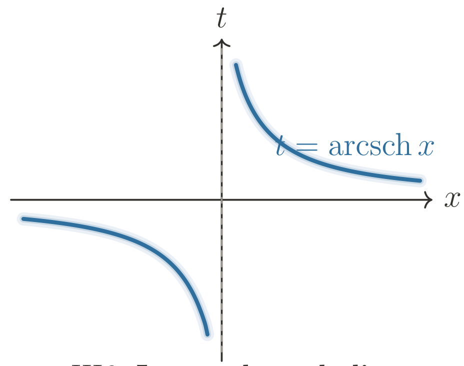


FIGURE HI6. **Inverse hyperbolic cosecant:**
Domain and range $\mathbb{R} \setminus \{0\}$; the graph keeps the positive and negative branches.

Definition

Definition 2.0.64 (Inverse hyperbolic cosecant)**.** The *inverse hyperbolic cosecant* function is the inverse of csch on its full domain. It is denoted by arcsch and has domain $\mathbb{R} \setminus \{0\}$ and range $\mathbb{R} \setminus \{0\}$.

$$\operatorname{arcsch} x = t \quad \Longleftrightarrow \quad \operatorname{csch} t = x \quad \text{and} \quad t \neq 0.$$

Remark (Domain and range). The regular function csch is undefined at 0. Its positive and negative branches are one-to-one, with ranges $(0, \infty)$ and $(-\infty, 0)$, so the inverse has domain and range $\mathbb{R} \setminus \{0\}$.

Remark (Dependencies).

- Hyperbolic cosecant

Theorem

Theorem 2.0.65 (Principal inverse hyperbolic cosecant identities)**.** *For every* $x \in \mathbb{R}\setminus\{0\}$,

$$\operatorname{csch}(\operatorname{arcsch} x) = x.$$

For every $t \in \mathbb{R} \setminus \{0\}$,

$$\operatorname{arcsch}(\operatorname{csch} t) = t.$$

Remark (Proof). The positive and negative branches of csch are one-to-one, and their ranges are disjoint. Therefore the two-branch inverse has the displayed cancellation laws.

Remark (Dependencies).

- Inverse hyperbolic cosecant

Theorem

Theorem 2.0.66 (Existence of hyperbolic sector points). *For every $t \geq 0$, there exists a point $P = (u, v)$ on the upper right branch of $x^2 - y^2 = 1$ with*

$$2\,\mathrm{HArea}(O, V, P) = t.$$

Remark (Proof). Parametrize the upper right branch by the x-coordinate:

$$P_x = (x, \sqrt{x^2 - 1}), \qquad x \geq 1.$$

Let

$$F(x) = 2\,\mathrm{HArea}(O, V, P_x).$$

The sector area varies continuously with x: the boundary point depends continuously on x, the function $\sqrt{x^2 - 1}$ is continuous on $[1, \infty)$, and the area swept over a compact interval changes continuously with the endpoint. Also $F(1) = 0$, since the sector degenerates at V. As $x \to \infty$, the swept sector area tends to infinity. Moreover, if $1 \leq x_1 < x_2$, the sector from V to P_{x_1} is properly contained in the sector from V to P_{x_2}, so $F(x_1) < F(x_2)$. Thus F is continuous, strictly increasing, begins at 0, and is unbounded. By the intermediate value property, every $t \geq 0$ is equal to $F(x)$ for some $x \geq 1$.

Remark (Dependencies).

- Hyperbolic sector

Theorem

Theorem 2.0.67 (Uniqueness of hyperbolic sector points). *For every $t \geq 0$, the point P in the existence theorem is unique.*

Remark (Proof). The function $F(x) = 2\,\mathrm{HArea}(O, V, P_x)$ is strictly increasing. Hence $F(x_1) = F(x_2)$ implies $x_1 = x_2$. Since the upper right branch point $P_x = (x, \sqrt{x^2 - 1})$ is determined by x, the point P is unique.

Remark (Dependencies).

- Existence of hyperbolic sector points

Corollary

Corollary 2.0.68 (Well-definedness of hyperbolic sine and cosine). *The assignments* $\cosh, \sinh : \mathbb{R} \to \mathbb{R}$ *in the definition of hyperbolic cosine and the definition of hyperbolic sine are well-defined functions.*

Remark (Proof). Existence supplies a sector point for each $t \geq 0$, and uniqueness makes its coordinates independent of choice. The even/odd extension then gives exactly one value for each negative parameter.

Remark (Dependencies).

- Existence of hyperbolic sector points
- Uniqueness of hyperbolic sector points

Theorem

Theorem 2.0.69 (Fundamental hyperbolic identity)**.** *For every $t \in \mathbb{R}$,*

$$\cosh^2 t - \sinh^2 t = 1.$$

Remark (Proof)*.* For $t \geq 0$, the point $P_t = (\cosh t, \sinh t)$ was chosen on the branch $x^2 - y^2 = 1$, so substituting its coordinates gives $\cosh^2 t - \sinh^2 t = 1$. For $t < 0$, use the extension $\cosh(-t) = \cosh t$ and $\sinh(-t) = -\sinh t$ to reduce to the nonnegative case.

Remark (Interpretation)*.* This identity justifies the word *hyperbolic*: it follows from membership on the unit hyperbola exactly as $\cos^2\theta + \sin^2\theta = 1$ follows from membership on the unit circle.

Remark (Dependencies).

- Hyperbolic sine
- Hyperbolic cosine

Theorem

Theorem 2.0.70 (Hyperbolic Pythagorean identity)**.** *For every $t \in \mathbb{R}$,*

$$\cosh^2 t - \sinh^2 t = 1.$$

Remark (Proof)*.* This is the fundamental hyperbolic identity.

Remark (Dependencies).

- Fundamental hyperbolic identity

Theorem

Theorem 2.0.71 (Exponential form of hyperbolic sine and cosine)**.** *For every $t \in \mathbb{R}$,*

$$\cosh t = \frac{e^t + e^{-t}}{2}, \qquad \sinh t = \frac{e^t - e^{-t}}{2}.$$

Remark (Proof)*.* Let $u = \cosh t$ and $v = \sinh t$. The identity $u^2 - v^2 = 1$ gives

$$(u - v)(u + v) = 1.$$

On the right branch, $u + v > 0$, so there is a unique real number s such that $u + v = e^s$. Then $u - v = e^{-s}$, and solving the two linear equations gives

$$u = \frac{e^s + e^{-s}}{2}, \qquad v = \frac{e^s - e^{-s}}{2}.$$

For the exponential parametrization

$$Q_s = \left(\frac{e^s + e^{-s}}{2}, \frac{e^s - e^{-s}}{2}\right),$$

a direct sector-area computation gives $2\,\mathrm{HArea}(O, V, Q_s) = s$. Therefore the defining sector parameter is $t = s$, and the displayed formulas follow. The negative case agrees with the even/odd extension.

Remark (Construction first). This theorem is derived from the geometric construction. It is not the definition of cosh and sinh in this chapter.

Remark (Dependencies).

- Hyperbolic sine
- Hyperbolic cosine
- Fundamental hyperbolic identity

Theorem

Theorem 2.0.72 (Hyperbolic sum formulas)**.** *For all $a, b \in \mathbb{R}$,*

$$\cosh(a+b) = \cosh a \cosh b + \sinh a \sinh b,$$

$$\sinh(a+b) = \sinh a \cosh b + \cosh a \sinh b.$$

Remark (Proof). Using the exponential form,

$$\cosh(a+b) = \frac{e^{a+b} + e^{-(a+b)}}{2}.$$

Expanding the right side of the claimed identity gives

$$\frac{e^a + e^{-a}}{2}\frac{e^b + e^{-b}}{2} + \frac{e^a - e^{-a}}{2}\frac{e^b - e^{-b}}{2} = \frac{e^{a+b} + e^{-(a+b)}}{2}.$$

The proof for $\sinh(a+b)$ is the same expansion with the signs arranged so that the mixed exponential terms survive with the claimed signs.

Remark (Dependencies).

- Exponential form of hyperbolic sine and cosine

Theorem

Theorem 2.0.73 (Hyperbolic double-angle formulas)**.** *For every $t \in \mathbb{R}$,*

$$\sinh(2t) = 2 \sinh t \cosh t,$$

$$\cosh(2t) = \cosh^2 t + \sinh^2 t = 2\cosh^2 t - 1 = 1 + 2\sinh^2 t.$$

Remark (Proof). Set $a = b = t$ in the hyperbolic sum formulas. The first two displayed formulas follow immediately. The remaining two forms of $\cosh(2t)$ follow by replacing $\sinh^2 t$ with $\cosh^2 t - 1$, or replacing $\cosh^2 t$ with $1 + \sinh^2 t$, using the hyperbolic Pythagorean identity.

Remark (Dependencies).

- Hyperbolic sum formulas
- Hyperbolic Pythagorean identity

Theorem

Theorem 2.0.74 (Hyperbolic tangent sum formula)**.** *For all $a, b \in \mathbb{R}$,*

$$\tanh(a+b) = \frac{\tanh a + \tanh b}{1 + \tanh a \tanh b}.$$

Remark (Proof). By definition,

$$\tanh(a+b) = \frac{\sinh(a+b)}{\cosh(a+b)}.$$

Substitute the sum formulas:

$$\frac{\sinh a \cosh b + \cosh a \sinh b}{\cosh a \cosh b + \sinh a \sinh b}.$$

Since $\cosh a$ and $\cosh b$ are nonzero, divide numerator and denominator by $\cosh a \cosh b$ to obtain the displayed identity.

Remark (Dependencies).

- Hyperbolic tangent
- Hyperbolic sum formulas

Theorem

Theorem 2.0.75 (Logarithmic form of inverse hyperbolic sine)**.** *For every $x \in \mathbb{R}$,*

$$\operatorname{arsinh} x = \log\left(x + \sqrt{x^2+1}\right).$$

Remark (Proof). Let $t = \operatorname{arsinh} x$. Then $x = \sinh t = (e^t - e^{-t})/2$. With $z = e^t > 0$, this becomes $z^2 - 2xz - 1 = 0$. The positive solution is $z = x + \sqrt{x^2+1}$, so $t = \log z$.

Remark (Dependencies).

- Inverse hyperbolic sine
- Exponential form of hyperbolic sine and cosine

Theorem

Theorem 2.0.76 (Logarithmic form of inverse hyperbolic cosine). *For every $x \in [1, \infty)$,*

$$\operatorname{arcosh} x = \log\left(x + \sqrt{x^2 - 1}\right).$$

Remark (Proof). Let $t = \operatorname{arcosh} x$. Then $t \geq 0$ and $x = \cosh t = (e^t + e^{-t})/2$. With $z = e^t \geq 1$, this becomes $z^2 - 2xz + 1 = 0$. The solution compatible with $z \geq 1$ is $z = x + \sqrt{x^2 - 1}$, so $t = \log z$.

Remark (Dependencies).

- Inverse hyperbolic cosine
- Exponential form of hyperbolic sine and cosine

Theorem

Theorem 2.0.77 (Logarithmic form of inverse hyperbolic tangent). *For every $x \in (-1, 1)$,*

$$\operatorname{artanh} x = \frac{1}{2} \log\left(\frac{1 + x}{1 - x}\right).$$

Remark (Proof). Let $t = \operatorname{artanh} x$. Then

$$x = \tanh t = \frac{e^t - e^{-t}}{e^t + e^{-t}} = \frac{e^{2t} - 1}{e^{2t} + 1}.$$

Solving gives $e^{2t} = (1 + x)/(1 - x)$, whose right side is positive because $-1 < x < 1$. Taking logarithms gives the formula.

Remark (Dependencies).

- Inverse hyperbolic tangent
- Exponential form of hyperbolic sine and cosine

Theorem

Theorem 2.0.78 (Logarithmic form of inverse hyperbolic cotangent). *For every $x \in (-\infty, -1) \cup (1, \infty)$,*

$$\operatorname{arcoth} x = \frac{1}{2} \log\left(\frac{x + 1}{x - 1}\right).$$

Remark (Proof). Let $t = \operatorname{arcoth} x$. Then

$$x = \coth t = \frac{e^t + e^{-t}}{e^t - e^{-t}} = \frac{e^{2t} + 1}{e^{2t} - 1}.$$

Solving gives $e^{2t} = (x + 1)/(x - 1)$. The quotient is positive exactly on the stated domain, and taking logarithms gives the formula.

Remark (Dependencies).

- Inverse hyperbolic cotangent
- Exponential form of hyperbolic sine and cosine

Theorem

Theorem 2.0.79 (Reciprocal form of inverse hyperbolic secant)**.** *For every* $x \in (0, 1]$,

$$\operatorname{arsech} x = \operatorname{arcosh}\left(\frac{1}{x}\right).$$

Remark (Proof). Let $t = \operatorname{arsech} x$. Then $t \geq 0$ and $x = \operatorname{sech} t = 1/\cosh t$, so $\cosh t = 1/x$. Since $0 < x \leq 1$, we have $1/x \in [1, \infty)$, and the definition of arcosh gives the displayed identity.

Remark (Dependencies).

- Inverse hyperbolic secant
- Inverse hyperbolic cosine

Theorem

Theorem 2.0.80 (Reciprocal form of inverse hyperbolic cosecant)**.** *For every* $x \in \mathbb{R} \setminus \{0\}$,

$$\operatorname{arcsch} x = \operatorname{arsinh}\left(\frac{1}{x}\right).$$

Remark (Proof). Let $t = \operatorname{arcsch} x$. Then $x = \operatorname{csch} t = 1/\sinh t$, so $\sinh t = 1/x$. The definition of arsinh gives the displayed identity.

Remark (Dependencies).

- Inverse hyperbolic cosecant
- Inverse hyperbolic sine

Remark (Forward pointer: behavior belongs in analysis). The continuity and differentiation results below are included here to keep the construction self-contained. Their canonical proofs belong after the limits, continuity, and differentiation machinery in the later analysis chapter on the behavior of elementary functions.

Theorem

Theorem 2.0.81 (Continuity of hyperbolic sine and cosine)**.** *The functions* sinh *and* cosh *are continuous on* $\mathbb{R}$.

Remark (Proof). By the exponential form,

$$\cosh t = \frac{e^t + e^{-t}}{2}, \qquad \sinh t = \frac{e^t - e^{-t}}{2}.$$

The exponential function is continuous, $t \mapsto -t$ is continuous, and sums, differences, scalar multiples, and compositions of continuous functions are continuous. Hence sinh and cosh are continuous on all of $\mathbb{R}$.

Remark (Dependencies).

- Exponential form of hyperbolic sine and cosine

Theorem

Theorem 2.0.82 (Continuity of the derived hyperbolic functions). *The functions* tanh *and* sech *are continuous on* $\mathbb{R}$*, and* coth *and* csch *are continuous on* $\mathbb{R} \setminus \{0\}$*.*

Remark (Proof). The functions tanh and sech are quotients of continuous functions by $\cosh t$, and $\cosh t \geq 1$, so the denominator never vanishes. The functions coth and csch are quotients by $\sinh t$, and $\sinh t = 0$ exactly at $t = 0$. The quotient rule for continuous functions gives the stated domains of continuity.

Remark (Dependencies).

- Hyperbolic tangent
- Hyperbolic secant
- Continuity of hyperbolic sine and cosine

Theorem

Theorem 2.0.83 (Derivatives of hyperbolic sine and cosine). *For every* $t \in \mathbb{R}$*,*

$$\frac{d}{dt} \sinh t = \cosh t, \qquad \frac{d}{dt} \cosh t = \sinh t.$$

Remark (Proof). Differentiate the exponential forms:

$$\frac{d}{dt} \sinh t = \frac{d}{dt} \left(\frac{e^t - e^{-t}}{2} \right) = \frac{e^t + e^{-t}}{2} = \cosh t,$$

and

$$\frac{d}{dt} \cosh t = \frac{d}{dt} \left(\frac{e^t + e^{-t}}{2} \right) = \frac{e^t - e^{-t}}{2} = \sinh t.$$

This uses linearity of differentiation and the chain rule applied to e^{-t}.

Remark (Dependencies).

- Exponential form of hyperbolic sine and cosine

Theorem

Theorem 2.0.84 (Derivative of hyperbolic tangent). *For every $t \in \mathbb{R}$,*

$$\frac{d}{dt} \tanh t = \operatorname{sech}^2 t = 1 - \tanh^2 t.$$

Remark (Proof). Use the quotient rule on $\tanh t = \sinh t / \cosh t$:

$$\frac{d}{dt} \tanh t = \frac{\cosh^2 t - \sinh^2 t}{\cosh^2 t} = \frac{1}{\cosh^2 t} = \operatorname{sech}^2 t.$$

The identity $\operatorname{sech}^2 t = 1 - \tanh^2 t$ follows by dividing $\cosh^2 t - \sinh^2 t = 1$ by $\cosh^2 t$.

Remark (Dependencies).

- Hyperbolic tangent
- Hyperbolic secant
- Derivatives of hyperbolic sine and cosine
- Hyperbolic Pythagorean identity

Theorem

Theorem 2.0.85 (Derivative of hyperbolic cotangent). *For every $t \neq 0$,*

$$\frac{d}{dt} \coth t = -\operatorname{csch}^2 t = 1 - \coth^2 t.$$

Remark (Proof). Use the quotient rule on $\coth t = \cosh t / \sinh t$:

$$\frac{d}{dt} \coth t = \frac{\sinh^2 t - \cosh^2 t}{\sinh^2 t} = -\frac{1}{\sinh^2 t} = -\operatorname{csch}^2 t.$$

The second form is obtained by dividing $\sinh^2 t - \cosh^2 t = -1$ by $\sinh^2 t$.

Remark (Dependencies).

- Hyperbolic cotangent
- Hyperbolic cosecant
- Derivatives of hyperbolic sine and cosine
- Hyperbolic Pythagorean identity

Theorem

Theorem 2.0.86 (Derivative of hyperbolic secant). *For every $t \in \mathbb{R}$,*

$$\frac{d}{dt} \operatorname{sech} t = -\operatorname{sech} t \, \tanh t.$$

Remark (Proof). Since $\operatorname{sech} t = (\cosh t)^{-1}$, the chain rule gives

$$\frac{d}{dt} \operatorname{sech} t = -(\cosh t)^{-2} \sinh t = -\frac{1}{\cosh t} \frac{\sinh t}{\cosh t} = -\operatorname{sech} t \, \tanh t.$$

Remark (Dependencies).

- Hyperbolic secant
- Hyperbolic tangent
- Derivatives of hyperbolic sine and cosine

Theorem

Theorem 2.0.87 (Derivative of hyperbolic cosecant). *For every* $t \neq 0$,

$$\frac{d}{dt} \operatorname{csch} t = -\operatorname{csch} t \, \coth t.$$

Remark (Proof). Since $\operatorname{csch} t = (\sinh t)^{-1}$, the chain rule gives

$$\frac{d}{dt} \operatorname{csch} t = -(\sinh t)^{-2} \cosh t = -\frac{1}{\sinh t} \frac{\cosh t}{\sinh t} = -\operatorname{csch} t \, \coth t.$$

Remark (Dependencies).

- Hyperbolic cosecant
- Hyperbolic cotangent
- Derivatives of hyperbolic sine and cosine

Proofs

Remark (Return). Return to Theorem

Theorem (Similarity invariance of trigonometric ratios). *If two right triangles have the same acute angle θ, then their corresponding opposite-to-hypotenuse and adjacent-to-hypotenuse ratios are equal.*

Professional Standard Proof.
TODO: professional standard proof for trig-ratios-similarity-invariant. ☐

Detailed Learning Proof.
TODO: detailed learning proof for trig-ratios-similarity-invariant. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: add dependencies for this proof.

Remark (Return). Return to Theorem

Theorem (Well-definedness of sine and cosine)**.** *For each acute Euclidean angle θ, the values $\sin\theta$ and $\cos\theta$ are independent of the right triangle chosen to compute them.*

Professional Standard Proof.
TODO: professional standard proof for sine-cosine-well-defined. ☐

Detailed Learning Proof.
TODO: detailed learning proof for sine-cosine-well-defined. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: add dependencies for this proof.

Remark (Return). Return to Theorem

Theorem (Six trigonometric functions)**.** *The six right-triangle trigonometric ratios define functions of an acute Euclidean angle, subject to denominator restrictions.*

Professional Standard Proof.
TODO: professional standard proof for six-trig-functions. □

Detailed Learning Proof.
TODO: detailed learning proof for six-trig-functions. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: add dependencies for this proof.

Remark (Return). Return to Theorem

Theorem (Unit-circle agreement with right-triangle trigonometry)**.** *For $0 < \theta < 90°$, the unit-circle definitions of sine and cosine agree with the right-triangle definitions.*

Professional Standard Proof.
TODO: professional standard proof for unit-circle-agrees-with-right-triangle-trig. □

Detailed Learning Proof.
TODO: detailed learning proof for unit-circle-agrees-with-right-triangle-trig. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: add dependencies for this proof.

Remark (Return). Return to Theorem

Theorem (Symmetry and periodicity of sine and cosine). *For every angle in the unit-circle domain, cosine is even, sine is odd, and one full turn returns the same sine and cosine values.*

Professional Standard Proof.
TODO: professional standard proof for trig-symmetry-periodicity. ☐

Detailed Learning Proof.
TODO: detailed learning proof for trig-symmetry-periodicity. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: add dependencies for this proof.

Remark (Return). Return to Theorem

Theorem (Pythagorean identity). *For every angle in the unit-circle domain,*

$$\cos^2\theta + \sin^2\theta = 1.$$

Professional Standard Proof.
TODO: professional standard proof for pythagorean-trig-identity. ☐

Detailed Learning Proof.
TODO: detailed learning proof for pythagorean-trig-identity. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: add dependencies for this proof.

Remark (Return). Return to Theorem

Theorem (Reciprocal identities)**.** *Where both sides are defined, secant, cosecant, and cotangent are the corresponding reciprocal ratios.*

Professional Standard Proof.
TODO: professional standard proof for trig-reciprocal-identities. ☐

Detailed Learning Proof.
TODO: detailed learning proof for trig-reciprocal-identities. ☐

Remark (Return). Return to Theorem

Theorem (Quotient identities). *Where both sides are defined, tangent and cotangent are the quotient ratios of sine and cosine.*

Professional Standard Proof.
TODO: professional standard proof for trig-quotient-identities. □

Detailed Learning Proof.
TODO: detailed learning proof for trig-quotient-identities. □

Remark (Return). Return to Theorem

Theorem (Tangent and secant Pythagorean identities). *Where both sides are defined,*

$$1 + \tan^2\theta = \sec^2\theta, \qquad 1 + \cot^2\theta = \csc^2\theta.$$

Professional Standard Proof.
TODO: professional standard proof for trig-tangent-pythagorean-identities. □

Detailed Learning Proof.
TODO: detailed learning proof for trig-tangent-pythagorean-identities. □

Remark (Return). Return to Theorem

Theorem (Angle-sum identities)**.** *The sine and cosine of a sum of angles are determined by the standard angle-sum formulas.*

Professional Standard Proof.
TODO: professional standard proof for trig-angle-sum-identities. □

Detailed Learning Proof.
TODO: detailed learning proof for trig-angle-sum-identities. □

Remark (Return). Return to Theorem

Theorem (Angle-difference identities)**.** *The sine and cosine of a difference of angles are determined by the standard angle-difference formulas.*

Professional Standard Proof.
TODO: professional standard proof for trig-angle-difference-identities. ☐

Detailed Learning Proof.
TODO: detailed learning proof for trig-angle-difference-identities. ☐

Remark (Return). Return to Theorem

Theorem (Double-angle identities). *The sine and cosine of twice an angle are determined by the standard double-angle formulas.*

Professional Standard Proof.
TODO: professional standard proof for trig-double-angle-identities. □

Detailed Learning Proof.
TODO: detailed learning proof for trig-double-angle-identities. □

Remark (Return). Return to Theorem

Theorem (Half-angle identities). *The sine and cosine of half an angle are determined by the standard half-angle formulas, with signs chosen by quadrant.*

Professional Standard Proof.
TODO: professional standard proof for trig-half-angle-identities. □

Detailed Learning Proof.
TODO: detailed learning proof for trig-half-angle-identities. □

Remark (Return). Return to Theorem

Theorem (Product-to-sum identities)**.** *Products of sine and cosine values can be rewritten as sums of sines or cosines.*

Professional Standard Proof.
TODO: professional standard proof for trig-product-to-sum-identities. □

Detailed Learning Proof.
TODO: detailed learning proof for trig-product-to-sum-identities. □

Remark (Return). Return to Theorem

Theorem (Sum-to-product identities)**.** *Sums and differences of sine and cosine values can be rewritten as products using midpoint and half-difference angles.*

Professional Standard Proof.
TODO: professional standard proof for trig-sum-to-product-identities. □

Detailed Learning Proof.
TODO: detailed learning proof for trig-sum-to-product-identities. □

Chapter 3

Analytical Geometry

SERIES
From Cantor to Ito

VOLUME I
Logic, Sets, and Proof

BOOK III
Foundational Geometry

CHAPTER 12
Analytical Geometry

3.1 Analytical Geometry

Remark (Status). Notes, proofs, and exercises will appear here in a future revision.

3.2 Lines

Definition

Definition 3.2.1 (Point-slope form). Let $P_1 = (x_1, y_1)$ be a point in the Euclidean plane and let $m \in \mathbb{R}$. The point-slope form of the line through P_1 with slope m is

$$y - y_1 = m(x - x_1).$$

Remark (Standard quantified statement). For every point $P_1 = (x_1, y_1) \in \mathbb{R}^2$ and every $m \in \mathbb{R}$, the corresponding line is the set of all $(x, y) \in \mathbb{R}^2$ satisfying $y - y_1 = m(x - x_1)$.

Remark (Derivation). The formula records the condition that the slope from P_1 to an arbitrary point (x, y) on the line is constant and equal to m.

Remark (Dependencies).

- Real Numbers
- Cartesian Product
- Subtraction on $\mathbb{R}$
- Multiplication on $\mathbb{R}$

Definition

Definition 3.2.2 (Two-point form)**.** Let $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ be distinct points with $x_1 \neq x_2$. The two-point form of the line through P_1 and P_2 is

$$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x - x_1).$$

Remark (Standard quantified statement)*.* For all distinct points $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ in $\mathbb{R}^2$ with $x_1 \neq x_2$, the line through P_1 and P_2 is the set of all $(x, y) \in \mathbb{R}^2$ satisfying the displayed two-point equation.

Remark (Dependencies)*.*

- Point-Slope Form
- Subtraction on $\mathbb{R}$
- $\mathbb{R}$ Is a Field

Definition

Definition 3.2.3 (General linear form)**.** The general linear form of a line in $\mathbb{R}^2$ is

$$Ax + By + C = 0,$$

where $A, B, C \in \mathbb{R}$ and $A^2 + B^2 \neq 0$.

Remark (Standard quantified statement)*.* For all $A, B, C \in \mathbb{R}$ with $A^2 + B^2 \neq 0$, the equation $Ax + By + C = 0$ determines a nondegenerate line in $\mathbb{R}^2$.

Remark (Interpretation)*.* The vector (A, B) is normal to the line. This form includes vertical lines and therefore avoids the exceptional cases present in slope-based forms.

Remark (Dependencies)*.*

- Real Numbers
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$
- $\mathbb{R}$ Is a Field

Definition

Definition 3.2.4 (Vector parametric form)**.** A line in $\mathbb{R}^2$ may be represented as the image of $\mathbf{r} : \mathbb{R} \to \mathbb{R}^2$ given by

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v},$$

where $\mathbf{r}_0 \in \mathbb{R}^2$ is a point on the line and $\mathbf{v} \in \mathbb{R}^2$ is a nonzero direction vector.

Remark (Standard quantified statement). For every $\mathbf{r}_0 \in \mathbb{R}^2$ and every nonzero $\mathbf{v} \in \mathbb{R}^2$, the set $\{\mathbf{r}_0 + t\mathbf{v} : t \in \mathbb{R}\}$ is a line in $\mathbb{R}^2$.

Remark (Dependencies).

- Real Numbers
- Cartesian Product
- Function
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$

Definition

Definition 3.2.5 (Intercept form)**.** If a line meets the x-axis at $(a, 0)$ and the y-axis at $(0, b)$ with $a, b \neq 0$, its intercept form is

$$\frac{x}{a} + \frac{y}{b} = 1.$$

Remark (Standard quantified statement). For all nonzero $a, b \in \mathbb{R}$, the line with intercepts $(a, 0)$ and $(0, b)$ is the set of all $(x, y) \in \mathbb{R}^2$ satisfying $x/a + y/b = 1$.

Remark (Derivation). The intercept form follows by applying the two-point form to the points $(a, 0)$ and $(0, b)$ and then rearranging the resulting equation.

Remark (Dependencies).

- Two-Point Form
- $\mathbb{R}$ Is a Field

Definition

Definition 3.2.6 (Affine parametric line in $\mathbb{R}^n$)**.** Given distinct points $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$, the line through them is

$$\mathbf{x}(t) = \mathbf{x}_1 + t(\mathbf{x}_2 - \mathbf{x}_1), \qquad t \in \mathbb{R}.$$

Remark (Standard quantified statement). For all distinct $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$, the set $\{\mathbf{x}_1 + t(\mathbf{x}_2 - \mathbf{x}_1) : t \in \mathbb{R}\}$ is the affine line through $\mathbf{x}_1$ and $\mathbf{x}_2$.

Remark (Interpretation). For $0 \leq t \leq 1$, the same formula describes the line segment joining $\mathbf{x}_1$ and $\mathbf{x}_2$. For arbitrary real t, it describes the entire affine line.

Remark (Dependencies).

- Real Numbers
- Cartesian Product

- Subtraction on $\mathbb{R}$
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$

Definition

Definition 3.2.7 (Normal form)**.** A line in $\mathbb{R}^2$ may be written in normal form as

$$x \cos \alpha + y \sin \alpha - p = 0,$$

where $p \geq 0$ is the distance from the origin to the line.

Remark (Standard quantified statement)*.* For every $p \geq 0$ and every real angle α, the equation $x \cos \alpha + y \sin \alpha - p = 0$ describes a line whose unit normal direction is $(\cos \alpha, \sin \alpha)$.

Remark (Dependencies)*.*

- General Linear Form
- Acute-angle sine
- Acute-angle cosine
- Order on $\mathbb{R}$

Definition

Definition 3.2.8 (Symmetric form)**.** Given a point (x_1, y_1) and direction cosines $(\cos \theta, \sin \theta)$, the symmetric form of a line is

$$\frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r.$$

Remark (Standard quantified statement)*.* For every point (x_1, y_1) and every direction with nonzero displayed denominators, the symmetric equation describes the same line as the parametric equations $x = x_1 + r \cos \theta$ and $y = y_1 + r \sin \theta$.

Remark (Dependencies)*.*

- Vector Parametric Form
- Acute-angle sine
- Acute-angle cosine
- $\mathbb{R}$ Is a Field

Proofs

Bibliography

Euclid. *The Thirteen Books of Euclid's Elements*. Dover Publications, 1956.